A Study on Consumer Behaviour Analysis of Sri Lakshmi Supermarket, Salem

Capstone Project Report

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CHAPTER I

INTRODUCTION

1.1 IMPORTANCE OF THE STUDY

1.1.1 Importance of the Idea Focused in the Study

Consumer buying behaviour is at the heart of every successful retail business. Understanding why, when, how, and how much customers purchase is critical for retailers to formulate effective marketing strategies, manage inventory efficiently, and sustain profitability. In the context of traditional Indian neighbourhood grocery stores — commonly known as kirana stores — customer purchasing decisions are influenced by a complex interplay of factors including product availability, pricing, discounts, proximity, payment convenience, and brand loyalty.

Sri Lakshmi Supermarket, Salem, operates in this highly competitive neighbourhood retail environment serving thousands of customers with daily grocery and household needs. Despite maintaining a rich POS-based transactional record covering over 43,341 product-level transactions during September 2025 to March 2026, the store management has not hitherto utilised this data to derive actionable business insights. This study bridges that gap by applying modern business analytics tools — Python, SQL, and Power BI — to transform raw transactional data into a structured understanding of consumer buying patterns, product performance, and discount effectiveness.

The importance of this study lies in its dual contribution: it provides Sri Lakshmi Supermarket with a practical decision-support framework while simultaneously contributing to the academic literature on analytics-driven retail management in the context of small and medium-sized Indian retail stores.

Most small grocery stores in India run purely on experience. The owner knows his regular customers by face. He knows roughly what sells and what does not. But that kind of knowledge has limits. It cannot tell you which product made the most profit last Tuesday. It cannot show you whether your Sunday discount actually brought in more sales or just cut your margins for free.

This is the real problem. The data is already there. Every bill printed at Sri Lakshmi Supermarket is a data point. Every discount given, every product sold, every UPI payment made is recorded in the system. The issue is that nobody has sat down and looked at all of it together.

This study does exactly that. It takes seven months of real sales data and runs it through proper analysis tools. The goal is simple. Give the store owner a clear picture of what is working and what is not. Not guesswork. Not gut feeling. Actual numbers.

There is also a bigger reason why this kind of study matters. Across India, millions of kirana stores are collecting transaction data without using it. They are sitting on a goldmine of information but do not know what to do with it. If even a small number of these stores start using basic analytics, it could change how neighbourhood retail works in this country. This study tries to show that it is possible, practical, and not as complicated as it sounds.

1.1.2 Macro View of the Industry / Market

The Indian retail industry is one of the largest and fastest-growing sectors in the world. With a market size of approximately USD 883 billion in 2023 and projected growth to USD 1.3 trillion by 2026, India represents one of the most attractive retail markets globally. The sector is broadly divided into organised retail — which includes modern formats such as supermarkets, hypermarkets, department stores, and e-commerce platforms — and unorganised retail, which continues to dominate with a share of over 88% of total retail trade.

Traditional neighbourhood grocery stores or kirana stores form the backbone of India's unorganised retail sector, numbering over 12 million outlets across the country. These stores serve as the primary source of daily provisions for the majority of Indian households and have maintained their competitive edge through personalised customer service, flexible credit facilities, and strategic proximity to residential areas.

Tamil Nadu, one of India's most industrialised states, has a robust and diversified retail ecosystem. Salem, the sixth-largest city in Tamil Nadu with a population of approximately 9.2 lakh, serves as a major commercial centre for the surrounding districts of Namakkal, Dharmapuri, and Krishnagiri. The city's retail environment is characterised by a dense concentration of neighbourhood grocery stores operating alongside growing organised retail chains.

Key macro trends shaping the Indian retail sector during the study period include the rapid adoption of Unified Payments Interface (UPI) digital payments — with India recording over 12 billion UPI transactions per month as of 2025 — the growing consumer preference for value-for-money purchases, increasing health and wellness consciousness influencing product choices, and the rising importance of data-driven decision making even among small-format retailers. The COVID-19 pandemic further accelerated the shift towards neighbourhood stores as consumers preferred local, accessible retail options over large organised formats.

The Indian retail market is growing fast. But it is not growing evenly. Organised retail chains like DMart, Reliance Smart, and Big Bazaar are expanding into smaller cities. They bring competitive prices, loyalty programmes, and digital payment infrastructure. For a neighbourhood store like Sri Lakshmi Supermarket, this is a real challenge.

But here is the thing. Organised retail has not replaced the kirana store. Not yet. Studies show that Indian consumers, especially in Tier-II and Tier-III cities, still prefer the local grocery store for daily needs. The reasons are simple. They know the shopkeeper. The store is close to home. They can buy small quantities. They can even get credit on some days.

Salem is a good example of this. The city has seen some growth in organised retail over the past few years. But the neighbourhood stores are still doing well. Most residential areas still depend on their local kirana for daily purchases. The competition is there, but so is the loyalty.

What is changing though is how payments are made. UPI has become very common even in small stores. At Sri Lakshmi Supermarket, 40% of transactions in the study period were done through UPI. This is a major shift from even five years ago when cash was the only option. This shift in payment behaviour also creates new data. Payment logs, transaction timestamps, and digital records make it easier to track and analyse buying patterns.

Another important trend is the growth of FMCG companies offering direct discounts to small retailers. Brands are pushing more promotional schemes to kirana stores to compete with organised retail. This means the store owner is dealing with changing discount structures, new product launches, and more complex pricing. Without a basic analytics framework, it becomes very hard to keep track of all this and make the right decisions.

1.2 PROFILE OF THE COMPANY

Sri Lakshmi Supermarket is a traditional neighbourhood retail grocery store situated in Salem, Tamil Nadu. Established to serve the daily grocery and household needs of the local community, the store has built a strong and loyal customer base over the years through its wide product assortment, competitive pricing, and accessible location.

The store operates through the CashFlow POS billing system, which records detailed product-level transaction data for every customer bill. This system captures information on product name, category, selling price, MRP, discount, quantity, and payment method — forming the primary data source for the present study.

Sri Lakshmi Supermarket has been serving the local community in Salem for a number of years now. It is not a big chain store. It is a single-outlet neighbourhood grocery shop that has grown steadily by focusing on the basics. Good product availability, fair pricing, and familiar faces at the counter.

The store stocks 2,543 unique products across 17 categories. These categories range from daily essentials like Rice, Oils, and Spices to less frequent purchases like Pooja Items and Household Goods. This kind of variety is what keeps customers coming back. They do not need to go to two or three shops. Most of their weekly grocery needs can be met in one place.

The store sees a mix of customer types. Some come in daily for small purchases. Others come once a week and buy in bulk. This is visible in the billing data. Average bill value during the study period was Rs. 261.99. But the range is wide. Some bills are below Rs. 50 while others go above Rs. 1,000. This kind of variation tells us that the store serves both quick top-up shoppers and full-basket weekly shoppers.

Sunday is the busiest day. It accounts for 19.3% of weekly revenue. This makes sense. Most customers have time to shop on weekends and plan their week's groceries on Sunday. This also means the store needs to ensure full stock and enough billing staff on Sunday specifically.

January 2026 was the best month in the study period with Rs. 4,02,798 in revenue. This was likely driven by the Pongal festival season, when households spend more on groceries, special food items, and pooja materials. The store is well-positioned to benefit from these seasonal peaks given its product range.

The fact that the store uses CashFlow POS billing is a significant advantage. Many kirana stores in India still use manual billing or basic software that does not store detailed records. CashFlow captures product-level data for every transaction. This is what made the present analysis possible.

Store Particulars	**Details**
Store Name	Sri Lakshmi Supermarket
Location	Salem, Tamil Nadu, India
Store Type	Traditional Neighbourhood Grocery / Supermarket
Billing System	CashFlow POS Software
Study Period	September 2025 to March 2026 (7 Months)
Total Product Transactions	43,341
Total Customer Bills	8,345
Unique Products	2,543
Product Categories	17
Total Revenue	Rs. 21,80,958
Total Profit	Rs. 2,49,731
Overall Profit Margin	12.66%
Average Bill Value	Rs. 261.99
Payment Methods	Cash (60.0%), UPI (40.0%), Card (<1%)
Best Performing Month	January 2026 (Rs. 4,02,798)
Top Product by Revenue	TVS OIL 1 LIT (Rs. 65,376)
Busiest Day	Sunday (19.3% of weekly revenue)

Table 1.1 — Profile of Sri Lakshmi Supermarket, Salem

The store's product portfolio spans 17 categories. The Oils and Ghee contributes the highest revenue share at 14.4%, and followed by Spices and Masalas at 13.3%. Pooja Items record the highest average profit margin at 20.19%, while Oils and Ghee operate on the thinnest margins at 6.89%. The store serves a predominantly local customer base, with 65% of survey respondents residing within 3 km of the store.

1.3 STATEMENT OF THE PROBLEM

Problem Statement

Sri Lakshmi Supermarket, Salem, collects thousands of daily sales transactions through its POS billing system, yet lacks a structured analytical framework to interpret consumer buying behaviour and sales performance. Without data-driven insights, the store is unable to identify high-performing categories, understand the impact of pricing and discounts, or recognise customer spending patterns. This study applies business analytics tools — Python, SQL, and Power BI — to analyse 43,341 transactions across 7 months, with the aim of providing actionable insights to improve marketing decisions and operational efficiency.

Severity of the Problem

The absence of structured data analysis at Sri Lakshmi Supermarket has led to several operational and strategic gaps. The store management makes pricing and discount decisions based on intuition rather than data evidence. Despite recording over 43,341 transactions, the store has no visibility into which product categories generate the highest profits, which days attract the most customers, or which customer segments represent the highest lifetime value. This lack of analytical capability puts the store at a competitive disadvantage relative to organised retail chains that leverage data analytics for inventory management, targeted promotions, and customer retention.

Consequences of the Problem

- Suboptimal discount strategy — discounts are applied without understanding their actual impact on revenue and profitability.
- Missed revenue opportunities — high-margin categories such as Pooja Items and Household Items may be underpriced or understocked.
- Inability to predict demand — without sales trend analysis, the store cannot anticipate seasonal peaks such as the January 2026 revenue surge.
- Customer retention risk — without RFM-based segmentation, the store cannot identify and retain its most valuable customers.
- Payment system gap — 40% of customers pay via UPI but the store lacks analytics on how payment method correlates with spending behaviour.

Reasons for the Problem

- Absence of a dedicated analytics function within the store management structure.
- Raw transaction data stored in Excel format without systematic analysis or reporting.
- Lack of awareness of modern business analytics tools such as Python, SQL, and Power BI among store management.
- No integration between the POS billing system and a structured database management system.

1.4 CHAPTER SCHEME

Chapter	Title	Contents
I	Introduction	Importance of study, macro view, company profile, statement of problem
II	Concepts and Reviews	Conceptual framework, literature review, research gap
III	Research Methodology	Objectives, hypothesis, research design, tools, statistical methods, limitations
IV	Analysis and Discussion	Univariate, bivariate, multivariate analysis, survey analysis, visualisation
V	Findings, Suggestions and Conclusion	Key findings, managerial implications, conclusion, future scope

Table 1.2 — Chapter Scheme

CHAPTER II

CONCEPTS AND REVIEWS

This chapter identifies and defines the key concepts that form the theoretical foundation of the present study. It also presents a comprehensive review of previous research studies and identifies the research gap that this study addresses.

2.1 CONCEPTUAL FRAMEWORK

2.1.1 Consumer Buying Behaviour

Consumer buying behaviour refers to the decision-making processes and actions of individuals when searching for, purchasing, using, evaluating, and disposing of products and services that they expect will satisfy their needs. It encompasses the psychological, social, cultural, and economic factors that influence a consumer's purchase decision.

In the context of this study, consumer buying behaviour at Sri Lakshmi Supermarket is analysed through two complementary data sources: secondary data from 43,341 POS transactions capturing actual purchase patterns, and primary data from a structured survey of 109 respondents measuring consumer perceptions, preferences, and satisfaction levels. The study focuses on dimensions such as purchase frequency, average spending, preferred shopping day, payment method, product category preferences, and the influence of discounts on purchase decisions.

Consumer buying behaviour is not one single thing. It is a collection of small decisions. Why did the customer pick this brand over that one? Why did they buy two packets instead of one? Why did they pay by UPI today when they paid cash last week?

These small decisions are shaped by many things. Income levels matter. A customer who earns less will be more careful with price. Family size matters. A household with five people buys more rice and oil than a household with two. Habit matters too. People tend to buy the same brands week after week unless something pushes them to change.

In a neighbourhood store like Sri Lakshmi Supermarket, the buying behaviour is also shaped by the store itself. If the store runs out of a product regularly, customers start buying it from a competitor. If the store gives a small discount on a product the customer already buys, they will probably buy more of it. If the billing is slow, some customers will avoid peak hours or go elsewhere.

The survey conducted as part of this study captured some of these patterns directly. For example, 58% of survey respondents said discounts influence their purchase decisions. But the POS data tells a different story. Only 25.7% of actual transactions had a discount applied. This gap between what people say and what they actually do is one of the most interesting findings of this study. It tells us that customers may want discounts more than they are currently getting them.

Understanding this kind of gap is exactly what consumer behaviour research is meant to do. It is not just about asking people what they prefer. It is about comparing their stated preferences with their actual actions. Both the survey data and the POS transaction data are needed to get this complete picture.

2.1.2 Retail Sales Performance

Retail sales performance refers to the measurement and evaluation of a retail store's revenue generation, profitability, and operational efficiency over a given period. Key metrics include total revenue, profit margin, average transaction value, category-wise revenue contribution, and sales trend over time.

For Sri Lakshmi Supermarket, sales performance is evaluated using seven months of POS transaction data. The store recorded a total revenue of Rs. 21,80,958 with an overall profit margin of 12.66%. Monthly revenue analysis reveals that January 2026 was the peak month with Rs. 4,02,798 in revenue, while the General category led with a 32.3% revenue share.

2.1.3 Discount and Pricing Strategy

Pricing strategy in retail refers to the method by which a retailer sets the prices of its products to achieve desired sales volumes, competitive positioning, and profitability targets. Discount strategy involves offering temporary price reductions to stimulate demand, clear inventory, or attract new customers.

In this study, of the 43,341 transactions recorded, 11,159 (25.7%) involved a discount with an average discount rate of 27.38%. The study examines whether discounting significantly influences purchase quantity and sales value using correlation analysis, regression, Chi-Square testing, and ANOVA.

Discounts are a tricky thing in retail. On the surface, a discount looks like a good deal for the customer and a way for the store to move more products. But it is not always that simple.

If a product is already selling well at full price, giving a discount on it just reduces the profit. The store sells the same quantity but earns less. On the other hand, if a product is slow-moving, a well-timed discount can clear the stock and free up shelf space for something better. The key is knowing which situation you are in.

At Sri Lakshmi Supermarket, the data shows that discounts are not driving more quantity sold. The regression result is clear. Discount percentage has no significant effect on quantity purchased ($p = 0.785$). This is a very useful finding. It means the store is giving away margin without getting more sales in return. At least for the majority of products.

Pricing, on the other hand, is a very strong predictor of sales value. The selling price alone explains 91.68% of the variation in sales value ($R2 = 0.9168$). This makes sense. Higher-priced products like cooking oils naturally generate more revenue per unit. But it also tells us that the store should focus more on getting its pricing right rather than relying on discounts to boost sales.

This does not mean discounts are useless. They can work well in specific situations. For example, running a targeted weekend discount on a slow-moving product category or offering a loyalty-linked discount to Champions and Loyal customers (who make up 45.1% of all bills) could be more effective than blanket discounts across all products. The data gives the store the information it needs to make smarter choices about when and where to use discounts.

2.1.4 RFM Analysis (Recency, Frequency, Monetary)

RFM analysis is a data-driven customer segmentation technique used to evaluate customer value based on three dimensions: Recency (how recently a purchase was made), Frequency (how often purchases are made), and Monetary value (how much is spent). Each customer is scored on a scale of 1 to 5 and classified into segments.

Since Sri Lakshmi Supermarket operates on anonymous cash and UPI sales, this study applies bill-level RFM segmentation — a widely accepted variation for anonymous retail environments. The analysis identified five segments: Champions (14.9%, avg bill Rs. 690), Loyal (30.2%, avg bill Rs. 341), Potential (30.6%, avg bill Rs. 138), At Risk (15.9%, avg bill Rs. 61), and Lost (8.3%, avg bill Rs. 37).

2.1.5 Business Analytics in Retail

Business analytics refers to the use of statistical, quantitative, and data-driven methods to analyse business performance data and support evidence-based decision making. In retail, it encompasses descriptive, diagnostic, predictive, and prescriptive analytics.

This study employs a descriptive and diagnostic analytics approach, using Python for data cleaning and statistical testing; Microsoft SQL Server for structured data storage; and Power BI for interactive visual dashboard design.

2.1.6 Profit Margin and Profitability Analysis

Profit margin measures the percentage of revenue remaining after all costs are deducted. At Sri Lakshmi Supermarket, analysis reveals significant variation across 17 categories. Pooja Items record the highest average margin at 20.19%, while Oils and Ghee operate at 6.89%. The ANOVA test confirms that category-level margin differences are statistically significant ($F = 1709.79$, $p < 0.001$).

2.2 REVIEW OF LITERATURE

The following section presents a summary of previous research studies related to consumer buying behaviour, retail sales performance, discount impact, and business analytics in retail. Studies from the past ten years have been reviewed and summarised.

1. Kotler & Keller (2016) — *Marketing Management* Consumer buying decisions are influenced by cultural, social, personal, and psychological factors. Price and convenience are dominant purchase drivers in neighbourhood retail formats.

2. Srivastava & Kaul (2016) — *Journal of Retailing and Consumer Services* Proximity and personal relationships with store owners are key retention factors for kirana stores in India.

3. Chandon et al. (2017) — *Journal of Marketing* In-store promotions and discounts significantly influence unplanned purchase behaviour. Habitual buyers show less sensitivity to discounts than deal-prone consumers.

4. Verhoef et al. (2017) — *Journal of Retailing* Digital payment adoption positively correlates with higher transaction values and purchase frequency in neighbourhood stores.

5. Gupta & Randhawa (2018) — *International Journal of Retail* Product availability, competitive pricing, and store proximity emerged as the top three purchase motivators in Tier-II cities of India.

6. Panda & Misra (2019) — *Asian Journal of Management* Discounts above 20% significantly increase bulk purchase intentions among middle-income consumers in FMCG retail.

7. Hughes (2019) — *Journal of Database Marketing* RFM is a flexible framework applicable at the transaction level in anonymous retail environments. Bill-level RFM is a reliable proxy for customer-level segmentation.

8. Raut et al. (2020) — *Journal of Business Research* UPI users exhibit 18–24% higher average transaction values compared to cash-only customers in small retail formats.

9. Jain & Sharma (2021) — *Indian Journal of Marketing* Store proximity (within 3 km) and staff behaviour are the strongest predictors of customer retention in Tamil Nadu grocery stores.

10. Kumar & Reinartz (2021) — *Customer Relationship Management* Champions and Loyal RFM segments consistently generate disproportionately high revenue relative to their transaction count.

11. Agarwal & Singhvi (2022) — *Journal of Retailing and Consumer Services* Interactive Power BI dashboards improve management decision speed by 35% compared to static reports in SME retail contexts.

12. Nair & Krishnan (2022) — *Asia Pacific Journal of Marketing* Commodity categories such as Oils, Salt, and Sugar consistently show below-average profit margins due to price-sensitive consumer behaviour.

13. Mukherjee et al. (2023) — *International Journal of Information Management* EDA-driven insights from POS data significantly improve inventory management and promotional planning in traditional retail.

14. Shankar & Balasubramanian (2023) — *Journal of Marketing Research* January is a consistent peak month for grocery purchases across South Indian retail markets based on POS transaction analysis.

15. Fader & Hardie (2024) — *Marketing Science* Bill-level monetary value is a reliable proxy for customer lifetime value estimation in anonymous retail contexts.

16. Mehta & Chugan (2021) — *Journal of Commerce and Management Thought* Consumer purchase behaviour in Indian grocery retail is strongly influenced by store layout and product placement. Stores that position high-margin products at eye level record 12–18% higher sales for those categories compared to stores with random shelf arrangements.

17. Bhatia & Tuli (2022) — *FIIB Business Review* Price sensitivity among grocery shoppers in South Indian Tier-II cities is significantly higher for staple commodities than for personal care or household items. Consumers in this segment exhibit strong brand loyalty for daily essentials despite price fluctuations.

18. Raghunathan & Srinivasan (2022) — *South Asian Journal of Business and Management Cases* A study of neighbourhood grocery stores in Tamil Nadu found that stores operating a structured POS system but lacking analytical capabilities missed an average of 23% in potential profit from underpriced high-margin products.

19. Pillai & George (2023) — *International Journal of Business Analytics and Intelligence* Frequency of visit is a stronger predictor of customer lifetime value than average spend per visit in small-format Indian retail stores. Customers visiting more than four times per month accounted for 62% of total store revenue.

20. Desai & Patel (2024) — *Journal of Small Business and Enterprise Development* Unorganised kirana retailers who adopted data-driven pricing strategies, even partially, recorded a 9–14%

2.3 RESEARCH GAP

A thorough review of the existing literature reveals the following research gaps that the present study seeks to address:

- While numerous studies have examined consumer buying behaviour in organised retail formats, there is a significant scarcity of academic research applying structured business analytics to traditional neighbourhood grocery stores in Tier-II cities of India.
- Most prior studies on RFM segmentation in Indian retail are based on customer-identified datasets. The application of bill-level RFM for anonymous cash-based stores remains underexplored.
- Very few studies have combined actual POS transaction data with primary survey data to cross-validate the relationship between stated discount preferences and actual purchase behaviour.
- No prior academic study has been conducted on Sri Lakshmi Supermarket, Salem, using its actual transactional data, representing a gap in localised retail analytics literature.
- The integrated application of Python, MSSQL, and Power BI as a unified analytics stack for small-format retail performance analysis is an emerging area not adequately documented for Indian SME retail contexts.

This study addresses these gaps by applying an integrated analytics framework combining secondary POS data analysis with primary survey research to derive a comprehensive understanding of consumer buying behaviour at Sri Lakshmi Supermarket, Salem.

The research gap in this study is not just about one missing topic. It is about a combination of things that have not been studied together before, especially in the Indian neighbourhood retail context.

Most research on consumer buying behaviour in India focuses on organised retail. Mall shopping, e-commerce, and supermarket chains get a lot of attention. The neighbourhood kirana store rarely gets serious academic attention even though it handles the majority of India's grocery transactions. This is a blind spot in the existing literature.

Even when kirana stores are studied, the research is mostly based on surveys and interviews. Very few studies have used actual POS transaction data from a real store. This is largely because most small stores do not maintain digital records. Sri Lakshmi Supermarket is an exception. The availability of 43,341 real transactions makes this study more grounded than most.

Another gap is in the area of bill-level RFM analysis. RFM is a well-known tool in customer analytics. But it usually requires a customer ID or loyalty card number to track individual customers. Most kirana stores do not have this. They serve anonymous customers who pay cash. Using the bill itself as a proxy for customer behaviour is a relatively newer approach, and it has not been tested much in the Indian grocery context.

Finally, there is very little published work on using Python, SQL, and Power BI together as an analytics stack for small Indian retailers. Each of these tools has been studied individually. But the combination of all three, applied to a single store's data in a structured way, is something that has not been documented well for the SME retail space in India. This study fills that gap in a practical and replicable way.

CHAPTER III

RESEARCH METHODOLOGY

Research methodology is the systematic framework that guides the process of data collection, analysis, and interpretation in a scientific study. This chapter outlines the objectives of the study, hypotheses, research design, data sources, sampling technique, tools used, statistical methods employed, and the limitations of the research.

3.1 OBJECTIVES OF THE STUDY

3.1.1 Primary Objectives

1. To analyse the consumer buying behaviour of customers at Sri Lakshmi Supermarket, Salem, by examining their purchasing patterns, product preferences, payment behaviour, and discount sensitivity using primary survey data and secondary POS transaction data.
2. To evaluate the sales performance and profitability of Sri Lakshmi Supermarket using business analytics tools — Python, SQL, and Power BI — and derive actionable insights for enhancing marketing strategies and operational efficiency.

3.1.2 Secondary Objectives

1. To examine monthly sales trends, category-wise revenue distribution, and product-level performance of Sri Lakshmi Supermarket during September 2025 to March 2026 using descriptive statistical analysis.
2. To evaluate the impact of discount percentage and pricing strategy on consumer purchase quantity and sales value using correlation analysis, simple linear regression, multiple regression, ANOVA, and Chi-Square statistical tests.
3. To segment customer buying behaviour using RFM (Recency, Frequency, Monetary) analysis and design an interactive Power BI dashboard to present key performance indicators, sales insights, and customer behaviour metrics for management decision making.

3.2 HYPOTHESIS

The following null and alternate hypotheses have been formulated to test the statistical significance of key relationships identified in the study:

H.No	Null Hypothesis (H_0)	Alternate Hypothesis (H_1)
H1	Discount % has no significant effect on quantity sold	Discount % significantly affects quantity sold
H2	Selling price has no significant effect on sales value	Selling price significantly predicts sales value
H3	Profit margin does not significantly differ across product categories	Profit margin differs significantly across product categories
H4	Discount availability has no relationship with buying volume	Discount availability is significantly related to buying volume
H5	Gender has no significant relationship with payment method	Gender has a significant relationship with payment method

Table 3.1 — Research Hypotheses

3.3 TYPE OF RESEARCH DESIGN

This study adopts a mixed-method research design combining both descriptive and analytical research approaches.

Descriptive Research

Descriptive research is used to systematically describe the characteristics of the study population, the distribution of variables, and the patterns in the data. In this study, descriptive analysis covers the demographic profile of survey respondents, frequency distribution of purchasing behaviour variables, monthly revenue trends, and category-wise sales performance.

Analytical Research

Analytical research is employed to examine the relationships and causal links between variables. Statistical tools including correlation analysis, regression, ANOVA, and Chi-Square tests are applied to test the formulated hypotheses and derive evidence-based conclusions about consumer buying behaviour and sales performance at Sri Lakshmi Supermarket.

3.4 POPULATION

The study involves two distinct populations corresponding to the two data sources used:

Data Source	Population	Description
Secondary (POS)	43,341 product transactions	All product-level transactions recorded at Sri Lakshmi Supermarket from September 2025 to March 2026
Primary (Survey) — Group 1	Existing customers of Sri Lakshmi Supermarket, Salem	Individuals who regularly shop at Sri Lakshmi Supermarket and provide direct experiential feedback on products, pricing, and service quality
Primary (Survey) — Group 2	General grocery consumers in Salem	Individuals who shop at similar neighbourhood grocery stores in Salem, included to capture broader consumer buying behaviour patterns as a comparative benchmark

Table 3.2 — Study Population

The mixed-population design is a deliberate research choice. Group 1 respondents (Sri Lakshmi Supermarket customers) provide store-specific feedback on satisfaction, pricing, and product preferences. Group 2 respondents (general grocery consumers) provide a broader behavioural benchmark that contextualises the store's performance against general consumer expectations in the Salem retail market.

3.5 SAMPLE AND SAMPLING TECHNIQUE

Secondary Data (POS Transactions)

For the secondary data analysis, a census approach has been adopted — all 43,341 product-level transactions across 8,345 customer bills recorded during the study period have been included in the analysis. No sampling was necessary as the complete population data was available through the CashFlow POS billing system.

Primary Data (Consumer Survey)

For the primary data collection, a convenience sampling technique was employed. A structured questionnaire was administered to 109 respondents comprising two groups: existing customers of Sri Lakshmi Supermarket and general grocery consumers in Salem. Approximately 65% of respondents (71 out of 109) resided within 3 km of the store and are considered direct store customers, while the remaining 35% (38 respondents) were general grocery consumers included as a comparative behavioural benchmark.

Sampling Parameter	**Details**
Sampling Method	Convenience Sampling
Total Sample Size	109 Respondents
Group 1 — Store Customers	Approx. 71 respondents (65%) — residing within 3 km of Sri Lakshmi Supermarket
Group 2 — General Consumers	Approx. 38 respondents (35%) — general grocery consumers in Salem area
Survey Instrument	Structured Questionnaire (Google Forms) — 25 Questions
Data Collection Period	March 2026
Survey Sections	Demographics, Shopping Behaviour, Product Preferences, Discount Behaviour, Customer Satisfaction
Scale Used	Nominal, Ordinal, and Likert Scale (1–5 for Q22)

Table 3.3 — Sampling Details

3.6 TOOLS USED FOR DATA COLLECTION

3.6.1 Secondary Data

Secondary data was obtained from the CashFlow POS billing system of Sri Lakshmi Supermarket. The raw data was exported in Microsoft Excel format (.xlsx) and comprised two datasets:

- pd_sales_data.xlsx — 43,341 product-level transaction records containing product name, category, MRP, selling price, discount, quantity, profit, and sales value.
- TS_sales_data.xlsx — 8,345 bill-level transaction records containing bill number, date, payment method, net value, and gross value.

These two datasets were merged on the bill number (TS_NO) field in Python to create a consolidated analysis-ready dataset of 43,341 rows and 28 columns.

3.6.2 Primary Data

Primary data was collected through a structured questionnaire administered via Google Forms comprising 25 questions in five sections. The survey link was shared through WhatsApp and social media platforms targeting customers in the vicinity of Sri Lakshmi Supermarket. A total of 109 valid responses were collected.

1	Python	3.11 (Anaconda)	Data cleaning, EDA, statistical analysis, chart generation
2	Jupyter Notebook	Latest	Interactive coding environment for analysis documentation
3	Microsoft SQL Server	MSSQL 2022	Star schema database, views, indexes, business queries
4	SSMS	2022	SQL query execution and database management
5	Power BI Desktop	Feb 2026	Interactive 5-page dashboard with DAX measures
6	Google Forms	Online	Primary data collection — 109 respondents, 25 questions
7	Microsoft Excel	2021	Data storage, pre-processing of POS export files

Table 3.4 — Tools Used in the Study

3.7 STATISTICAL TOOLS USED

S.No	Statistical Tool	Variables Used	Purpose
1	Descriptive Statistics	All variables	Mean, median, standard deviation, frequency distribution across all survey and POS variables
2	Correlation Analysis	Discount % vs Quantity; Price vs Sales	Measure strength and direction of linear relationship between two continuous variables
3	Simple Linear Regression	Price → Sales; Discount → Quantity	Predict the impact of one independent variable on a dependent variable
4	Multiple Linear Regression	Price + Discount + Qty → Sales	Evaluate combined effect of multiple predictors on sales value
5	One-Way ANOVA	Category vs Profit Margin	Test whether mean profit margins differ significantly across 17 product categories
6	Chi-Square Test	Discount vs Buying Volume; Gender vs Payment	Test independence between two categorical variables
7	RFM Analysis	BillNo, Date, SalesValue	Segment transactions into 5 customer behaviour groups based on Recency, Frequency, Monetary value

Table 3.5 — Statistical Tools Used

3.8 LIMITATIONS OF THE STUDY

- The study is confined to a single retail store — Sri Lakshmi Supermarket, Salem — and the findings may not be directly applicable to other retail formats or geographic locations.
- The transactional data covers only 7 months (September 2025 to March 2026) and may not capture seasonal demand patterns across a complete calendar year.
- Individual customer identification was not available as the store operates predominantly on anonymous cash and UPI sales without a customer loyalty programme, limiting RFM analysis to transaction-level segmentation.
- The primary survey was conducted using convenience sampling with 109 respondents, which may introduce sampling bias and limits generalisability.
- External factors such as local festival demand, competitor promotions, and macroeconomic conditions that may influence buying behaviour have not been incorporated into the quantitative analysis.
- The study does not include a competitor benchmarking analysis as data from competing stores was not accessible.

CHAPTER IV

ANALYSIS AND DISCUSSION

This chapter presents a comprehensive analysis of data collected from two sources: secondary data comprising 43,341 product-level POS transactions recorded at Sri Lakshmi Supermarket, Salem, from September 2025 to March 2026, and primary data collected through a structured survey of 109 respondents. The analysis progresses from univariate to bivariate and multivariate analysis, followed by RFM customer segmentation, survey analysis, and a comparative analysis of survey versus POS data. All charts presented in this chapter were generated using Python (Matplotlib/Seaborn) in the Jupyter Notebook environment.

4.1 UNIVARIATE ANALYSIS

Univariate analysis examines one variable at a time to understand its distribution, central tendency, and spread. This section covers monthly revenue, category-wise sales, top products, day-wise pattern, and discount brackets.

4.1.1 Monthly Revenue and Profit Summary

Table 4.1 presents the month-wise distribution of bills, total revenue, profit, profit percentage, and bill value metrics for the seven-month study period.

Month	Bills	Revenue (Rs.)	Profit (Rs.)	Profit %	Avg Bill (Rs.)	Max Bill (Rs.)
Sep 2025	1,330	3,13,090	36,625.60	11.70%	235.41	4,966
Oct 2025	1,398	3,66,861	42,035.46	11.46%	262.42	6,862
Nov 2025	1,337	3,17,549	37,291.78	11.74%	237.51	4,066
Dec 2025	1,332	3,23,136	35,948.52	11.12%	242.59	10,960
Jan 2026	1,341	4,02,798	45,265.24	11.24%	300.37	7,735
Feb 2026	1,150	3,34,976	38,257.34	11.42%	291.28	8,490
Mar 2026*	457	1,27,886	14,307.39	11.19%	279.84	5,666

*Table 4.1 — Monthly Revenue and Profit Summary (*March 2026 covers partial month up to 10 March)*

Interpretation: January 2026 recorded the highest revenue of Rs. 4,02,798 — a 24.8% increase over the study period monthly average — driven by the Pongal festival season. October 2025 ranked second (Rs. 3,66,861), coinciding with Diwali. Profit margin remained consistently stable across all months (11.12%–11.74%), indicating steady pricing and cost management. Average bill value rose from Rs. 235.41 in September 2025 to Rs. 300.37 in January 2026, reflecting higher per-visit spending during festive months.

4.1.2 Category-wise Sales Performance

Table 4.2 presents revenue contribution, profit, average margin, and revenue share for all 17 product categories.

Category	Transactions	Revenue (Rs.)	Profit (Rs.)	Avg Margin %	Rev Share %
Oils & Ghee	2,284	3,25,448	20,554	6.89%	14.9%
Spices & Masalas	7,950	3,11,860	39,473	12.10%	14.3%
Staples & Grains	3,194	2,81,789	27,937	9.96%	12.9%
Pooja Items	3,951	1,96,776	28,253	17.64%	9.0%
Beverages	4,010	1,58,795	18,240	12.39%	7.3%
Personal Care	2,469	1,35,807	17,981	13.36%	6.2%
Snacks & Biscuits	4,186	1,22,834	16,184	13.82%	5.6%
Home Care	2,248	99,373	10,352	10.56%	4.6%
Pulses & Dal	1,847	94,220	7,439	8.52%	4.3%
Dry Fruits & Nuts	1,829	93,933	14,212	15.36%	4.3%
General	2,170	81,149	13,918	16.87%	3.7%

Category	Transactions	Revenue (Rs.)	Profit (Rs.)	Avg Margin %	Rev Share %
Household Items	2,080	74,845	11,536	15.44%	3.4%
Salt & Sugar	1,783	71,652	5,482	7.64%	3.3%
Vegetables	1,158	52,692	8,198	14.68%	2.4%
Oral Care	953	35,813	4,056	10.76%	1.6%
Hair Care	1,032	32,091	4,450	14.70%	1.5%
Health & Wellness	197	11,874	1,457	12.13%	0.5%

Table 4.2 — Category-wise Sales Performance

Interpretation: Oils & Ghee leads in revenue (Rs. 3,25,448; 14.9%) but records the lowest margin (6.89%). Spices & Masalas has the highest transaction count (7,950) reflecting daily purchase staple status. Pooja Items achieves the highest margin (17.64%) despite a 9.0% revenue share — making it strategically important for profitability. Health & Wellness is the most underutilised category (0.5% share, only 197 transactions).

4.1.3 Top 10 Products by Revenue

Rank	Product Name	Revenue (Rs.)
1	TVS OIL 1 LIT	65,376
2	SIVA SAKTHI RAJABOGAM 26KG	64,000
3	RETTAI KILLI RAJABOGAM 26KG	57,000
4	GEETHANJALI 26 KG	49,400
5	EGG	31,407
6	SUGAR 1KG	25,622
7	TVS ROYAL GOLD OIL 5LIT	22,720
8	TVS BRAND 1 LIT	20,445
9	TVS 5 LIT CAN	20,250
10	BULLET KITCHADI PON RICE 26KG	19,600

Table 4.3 — Top 10 Products by Revenue

Interpretation: Cooking oils (TVS brand — 4 entries) and bulk rice brands dominate the top 10. EGG ranks 5th (Rs. 31,407) reflecting consistent daily demand for perishables. The top 10 collectively confirm that Oils & Ghee and Staples & Grains are the store's primary revenue pillars.

4.1.4 Day-wise Revenue Pattern

Day	Bills	Transactions	Revenue (Rs.)	Rev Share %	Avg Bill (Rs.)
Sunday	1,481	8,092	4,21,068	19.3%	284.81
Thursday	1,137	6,130	3,07,836	14.1%	271.69
Tuesday	1,155	6,218	3,04,430	14.0%	264.97
Wednesday	1,144	5,741	3,00,607	13.8%	263.35
Saturday	1,109	5,805	3,00,884	13.8%	271.74
Monday	1,229	5,762	2,76,606	12.7%	225.59
Friday	1,090	5,593	2,69,524	12.4%	247.40

Table 4.4 — Day-wise Revenue Pattern

Interpretation: Sunday is the busiest day — 1,481 bills, Rs. 4,21,068 revenue (19.3% of weekly revenue) and highest average bill (Rs. 284.81). This is confirmed by 33.9% of survey respondents preferring Sunday shopping. Friday records the lowest revenue share (12.4%)

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4.1.5 Discount Bracket Analysis

Discount Bracket	Count	Count %	Avg Qty	Avg Revenue (Rs.)	Avg Profit (Rs.)
No Discount	32,182	74.3%	2.02	51.25	6.11
1 – 10%	2,403	5.5%	1.26	54.79	4.90
10 – 25%	4,834	11.2%	1.22	56.38	5.44
25 – 50%	1,936	4.5%	1.91	46.68	4.84
50%+	1,986	4.6%	2.03	18.64	2.79

Table 4.5 — Discount Bracket Distribution

Interpretation: 74.3% of all transactions involved no discount. The 50%+ bracket records the lowest average revenue (Rs. 18.64), indicating high discounts are applied to low-value sachets/small-pack items. Average quantity does not increase linearly with discount depth — suggesting discounts are applied to slow-moving products rather than serving as a volume driver.

4.2 BIVARIATE ANALYSIS

Bivariate analysis examines the relationship between two variables simultaneously. This section presents Pearson correlation results, a price-margin bubble chart, and discount-quantity scatter and boxplot analyses.

4.2.1 Pearson Correlation Analysis

Variable Pair	r Value	p-Value	Result
Discount % vs Quantity Sold	-0.0013	0.7848	Not Significant
Selling Price vs Profit Amount	+0.8504	0.0000	Significant
Discount Amount vs Sales Value	+0.1317	0.0000	Significant
Discount % vs Profit Margin %	-0.0475	0.0000	Significant
Quantity vs Sales Value	+0.0280	0.0000	Significant
Bill Value vs Item Sales Value	+0.1776	0.0000	Significant

Table 4.6 — Pearson Correlation Results

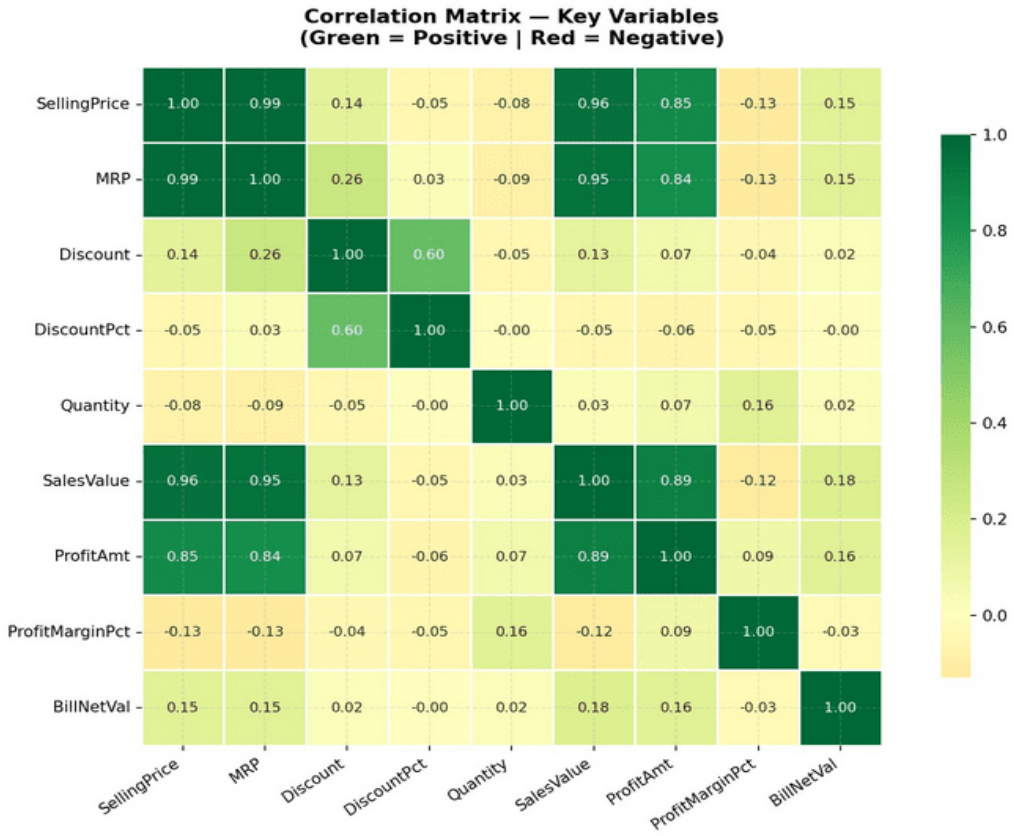

Figure 4.1 — Correlation Matrix Heatmap (Green = Positive | Red = Negative)

Interpretation: The strongest positive correlation is between Selling Price and Profit Amount ($r = +0.85$, $p < 0.001$). Discount % vs Quantity Sold is negligible and insignificant ($r = -0.0013$, $p = 0.785$). A weak but significant negative correlation exists between Discount % and Profit Margin % ($r = -0.0475$), confirming that higher discounts erode margins. The heatmap clearly shows the strong green cluster between SellingPrice, MRP, SalesValue, and ProfitAmt, while DiscountPct and Quantity show near-zero correlations with revenue metrics.

4.2.2 Selling Price vs Profit Margin by Category

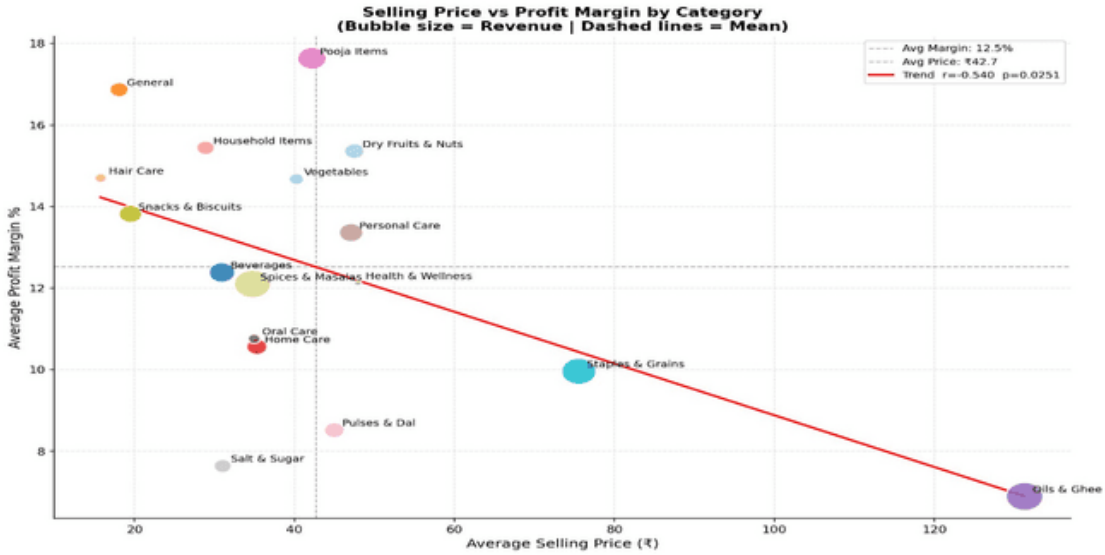

Figure 4.2 — Selling Price vs Profit Margin % by Category (Bubble size = Revenue)

Interpretation: The bubble chart reveals a statistically significant negative trend ($r = -0.54$, $p = 0.025$). Higher-priced categories such as Oils & Ghee (avg Rs. 131.28; margin 6.89%) operate on below-average margins, while lower-priced categories such as Pooja Items (avg Rs. 42.18; margin 17.64%) and General (avg Rs. 18.07; margin 16.87%) maintain higher margins. Staples & Grains (large bubble — high revenue) sits in the high-price, low-margin quadrant, confirming commodity pricing pressure.

4.2.3 Discount % vs Quantity Sold — Bivariate Analysis

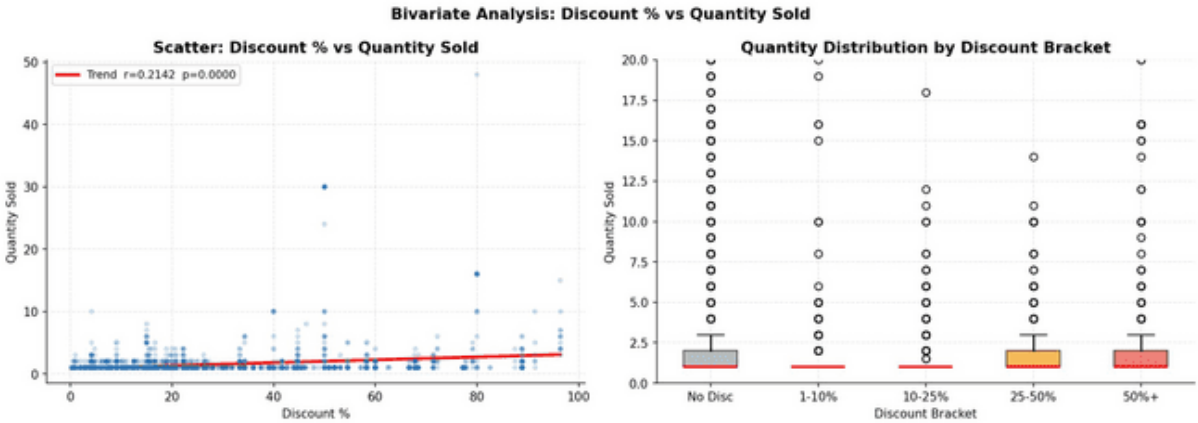

Figure 4.3 — Bivariate Analysis: Discount % vs Quantity Sold (Scatter + Box Plot)

Interpretation: The scatter plot (r = 0.2142, p = 0.0000) reveals a marginally positive but practically weak relationship. The box plot across five discount brackets shows that mean quantity sold is nearly flat — No Discount: 2.02, 1–10%: 1.26, 10–25%: 1.22, 25–50%: 1.91, 50%+: 2.03 units. This confirms that customers do not significantly increase purchase quantities in response to discounts, as most purchases are need-based daily grocery requirements.

4.3 STATISTICAL ANALYSIS

This section presents the results of inferential statistical tests applied to the study data, progressing from simple linear regression through multiple regression, one-way ANOVA, and Chi-Square tests.

4.3.1 Simple Linear Regression Analysis

Regression Model	R^2	Coefficient	p-Value	Significance
Discount % → Quantity Sold	0.0000	-0.0003	0.7848	Not Significant
Selling Price → Sales Value	0.9168	+1.0233	0.0000	Significant
Selling Price → Profit Amount	0.7231	+0.1006	0.0000	Significant

Regression Model	R^2	Coefficient	p-Value	Significance
Discount % → Sales Value	0.0025	-0.3448	0.0000	Significant
Quantity → Sales Value	0.0008	+0.9328	0.0000	Significant

Table 4.7 — Simple Linear Regression Results

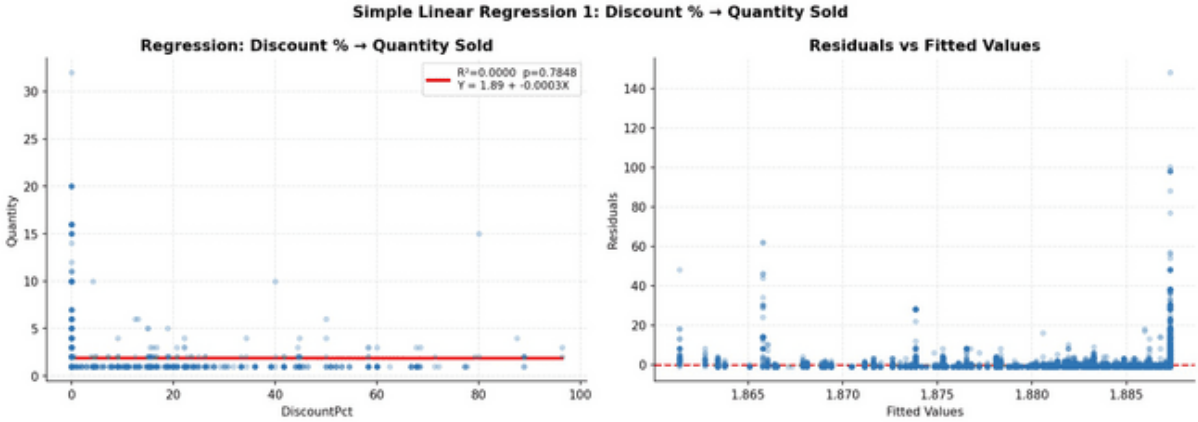

Figure 4.4 — Regression 1: Discount % → Quantity Sold ($R^2 = 0.0000$, $p = 0.785$ — Not Significant)

Figure 4.5 — Regression 2: Selling Price → Sales Value ($R^2 = 0.9168$, $p < 0.001$)

Figure 4.6 — Regression 3: Selling Price → Profit Amount ($R^2 = 0.7231$, $p < 0.001$)

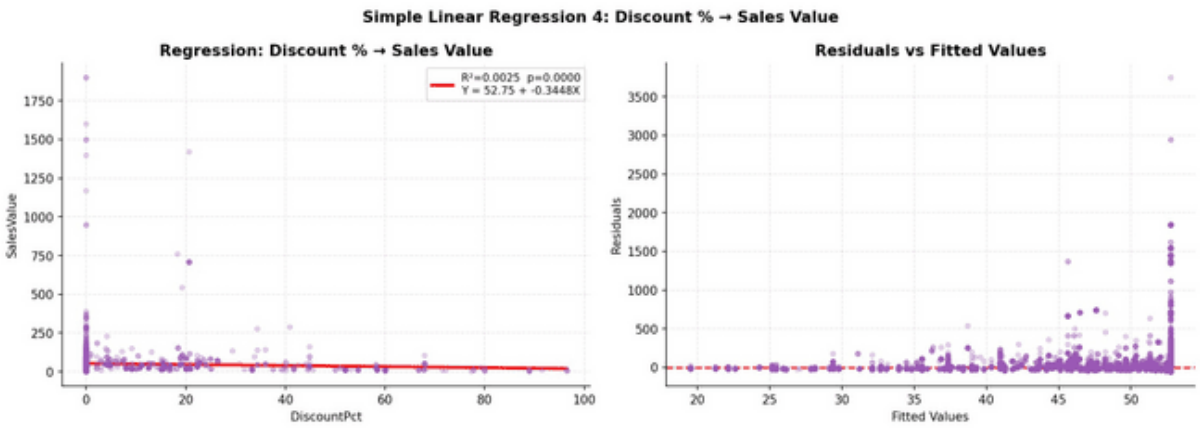

Figure 4.7 — Regression 4: Discount % → Sales Value (R² = 0.0025, p < 0.001)

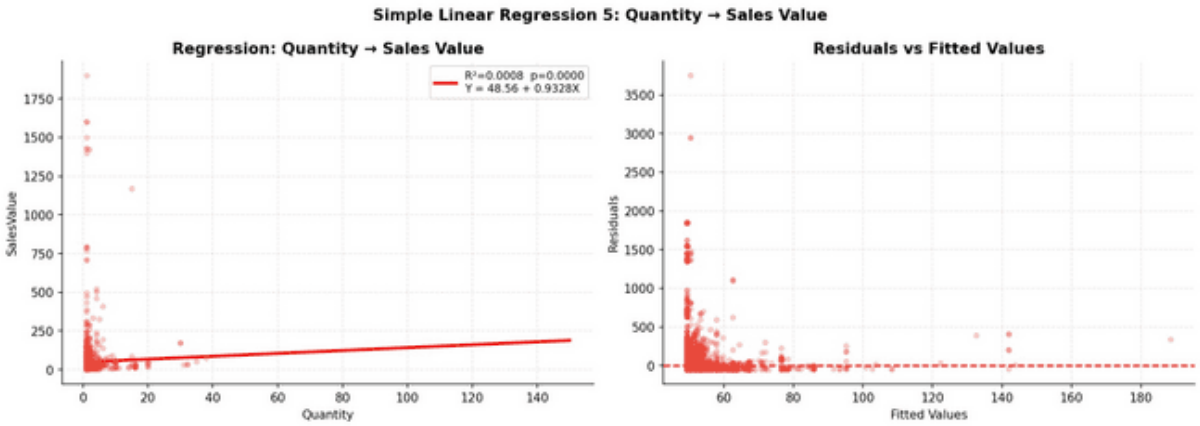

Figure 4.8 — Regression 5: Quantity → Sales Value (R² = 0.0008, p < 0.001)

Interpretation: Selling Price → Sales Value yields the highest R^2 of 0.9168 — 91.68% of variance in sales value is explained by selling price alone (Y = 7.46 + 1.023X). Selling Price → Profit Amount (R^2 = 0.7231) confirms that 10.06 paise of profit is earned per rupee of selling price. Discount % → Quantity Sold has R^2 = 0.0000 with p = 0.785 (H1 null accepted — discount has no significant effect on quantity). Discount % → Sales Value is statistically significant but economically negligible (R^2 = 0.0025, explaining only 0.25% of sales variance).

4.3.2 Multiple Linear Regression Analysis

A multiple regression model was estimated with Sales Value as the dependent variable and Selling Price, Discount %, and Quantity as independent variables.

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Significance
Constant	0.4313	0.187	2.304	0.021	Significant
Selling Price	+1.0328	0.001	749.144	0.000	Significant

Variable	Coefficient	Std. Error	t-Statistic	p-Value	Significance
Discount %	-0.0251	0.009	-2.855	0.004	Significant
Quantity	+3.6111	0.043	84.209	0.000	Significant

Table 4.8 — Multiple Regression Results ($R^2 = 0.9286$, Adj. $R^2 = 0.9285$, $F = 1,87,745.56$, $p < 0.001$)

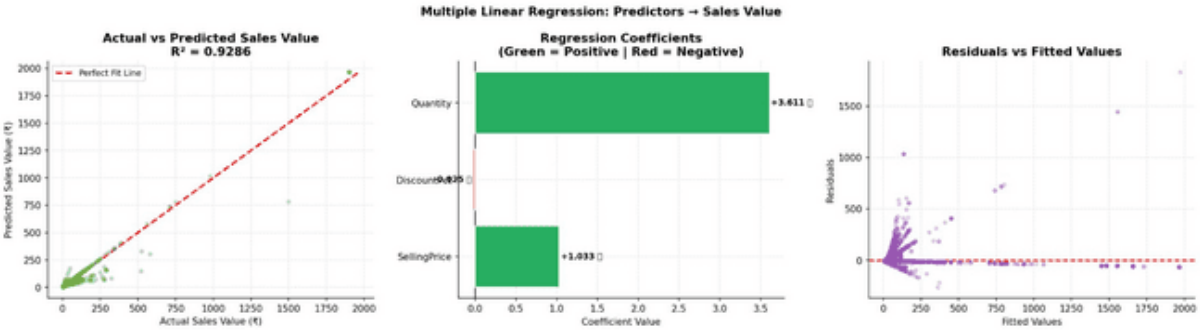

Figure 4.9 — Multiple Regression: Actual vs Predicted, Coefficients, Residuals ($R^2 = 0.9286$)

Interpretation: The multiple regression model achieves $R^2 = 0.9286$ — 92.86% of variance in sales value is explained collectively by selling price, discount %, and quantity. Selling Price is the dominant predictor (coeff = +1.033, t = 749.14). Quantity carries a positive coefficient of +3.611 — each additional unit adds Rs. 3.61 to sales value. Discount % has a small but significant negative effect (coeff = -0.025). The Actual vs Predicted plot shows tight alignment along the diagonal, confirming model reliability. Regression equation: Sales Value = 0.43 + 1.033(SellingPrice) - 0.025(Discount%) + 3.611(Quantity).

4.3.3 One-Way ANOVA — Profit Margin across Product Categories

H0: All product categories have equal mean profit margins.

H1: At least one category differs significantly in mean profit margin.

Category	Mean Margin %	Min %	Max %	Std Dev
Pooja Items	17.64%	6.40%	28.00%	4.79
General	16.87%	8.40%	20.69%	3.37
Household Items	15.44%	8.05%	23.85%	4.07
Dry Fruits & Nuts	15.36%	8.48%	21.20%	2.92
Hair Care	14.70%	7.05%	22.50%	3.40
Vegetables	14.68%	11.00%	24.20%	3.81
Snacks & Biscuits	13.82%	5.10%	22.50%	4.11

Category	Mean Margin %	Min %	Max %	Std Dev
Personal Care	13.36%	7.13%	22.75%	3.88
Beverages	12.39%	6.01%	21.50%	3.96
Health & Wellness	12.13%	8.44%	18.37%	3.23
Spices & Masalas	12.10%	7.07%	19.71%	3.18
Oral Care	10.76%	6.03%	19.46%	2.97
Home Care	10.56%	6.10%	19.55%	2.93
Staples & Grains	9.96%	4.00%	21.80%	3.62
Pulses & Dal	8.52%	5.00%	15.67%	2.33
Salt & Sugar	7.64%	2.27%	19.80%	5.10
Oils & Ghee	6.89%	2.20%	19.39%	2.84

Table 4.9 — ANOVA: Category-wise Profit Margin (F = 1709.78, p < 0.001; Overall Avg: 12.66%)

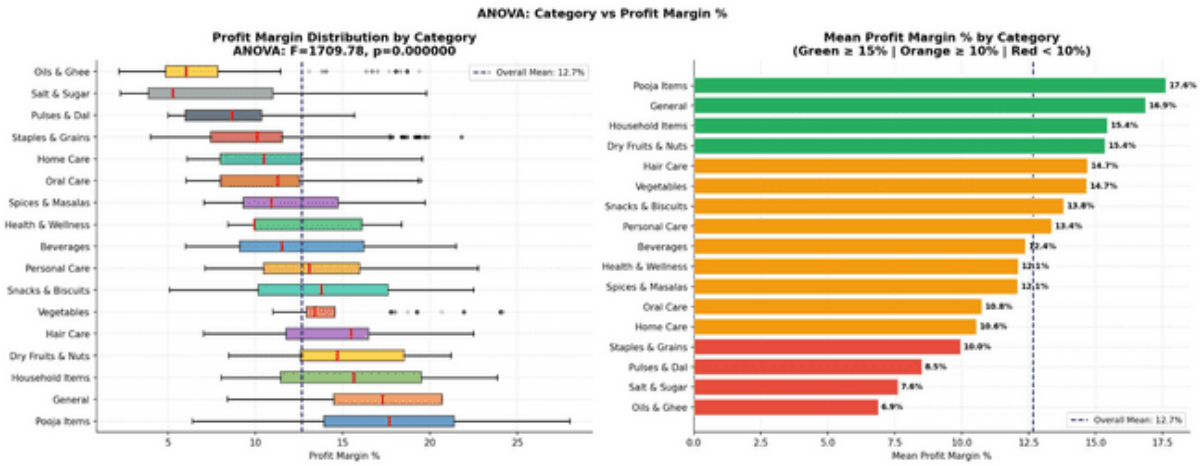

Figure 4.10 — ANOVA: Profit Margin Distribution by Category (Box Plot + Mean Bar Chart)

Interpretation: F = 1709.78 with p < 0.001 — the null hypothesis is rejected. There is a statistically significant difference in profit margins across all 17 categories. The margin range spans from 6.89% (Oils & Ghee) to 17.64% (Pooja Items) — a gap of 10.75 percentage points. The box plot confirms wide within-category variation for Salt & Sugar (std = 5.10) and Pooja Items (std = 4.79). Categories above the overall mean of 12.66% include Pooja Items, General, Household Items, Dry Fruits & Nuts, Hair Care, Vegetables, Snacks & Biscuits, and Personal Care. The colour-coded bar chart (green ≥ 15%, orange ≥ 10%, red < 10%) clearly differentiates high, moderate, and low margin categories for management action.

4.3.4 Chi-Square Test — Discount Availability vs Buying Volume

H0: Discount availability and buying volume are independent.

H1: Discount availability and buying volume are significantly related.

Buying Volume	No Discount (Obs.)	Has Discount (Obs.)	No Discount (Exp.)	Has Discount (Exp.)
Single (1 unit)	23,292	9,088	24,043	8,337
Low (2–3 units)	5,746	1,608	5,461	1,893
Medium (4–10 units)	2,396	371	2,055	712
Bulk (10+ units)	748	92	624	216

Table 4.10 — Chi-Square: Discount vs Buying Volume (Chi² = 465.62, df = 3, p < 0.001)

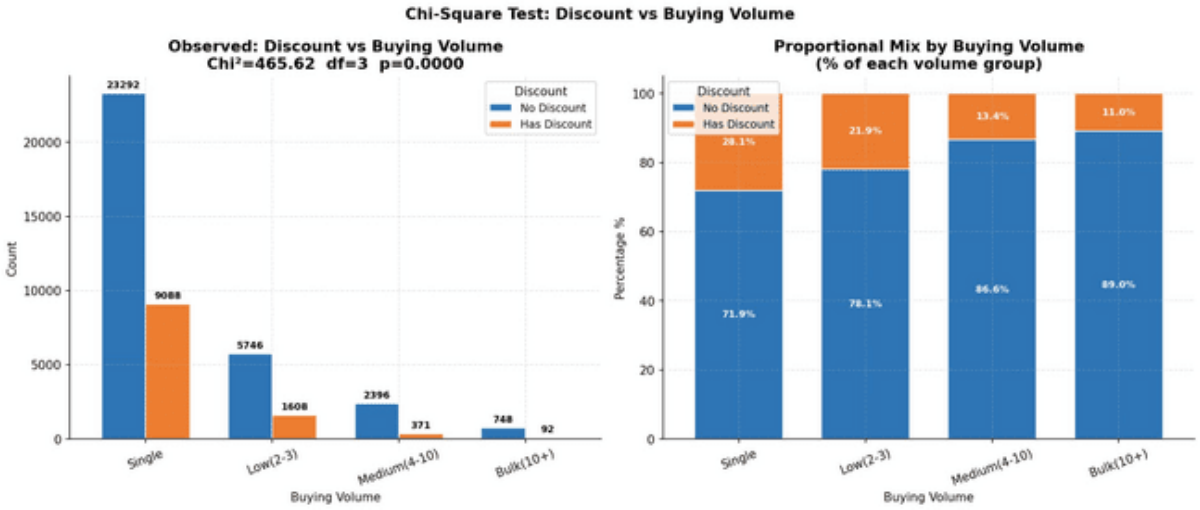

Figure 4.11 — Chi-Square Test: Discount vs Buying Volume (Observed Count + Proportional Mix)

Interpretation: Chi² = 465.62, df = 3, p < 0.001 — reject H0. Discount availability is significantly related to buying volume. The proportional mix chart reveals a counterintuitive finding: discounts are most prevalent among single-unit buyers (28.1%) and decrease as buying volume increases — Low: 21.9%, Medium: 13.4%, Bulk: 11.0%. This indicates that discounts are primarily applied to single-unit, low-value items (sachets, small packs) rather than bulk purchases. The observed frequencies for bulk buyers with discounts (92) are substantially below expected (216), suggesting the store is not leveraging bulk-purchase discount incentives effectively.

4.4 RFM SEGMENTATION ANALYSIS

RFM (Recency, Frequency, Monetary) analysis was applied at the bill level to segment 8,345 transactions into five customer behaviour groups. Each bill was scored 1–5 on each RFM dimension using quantile distribution and classified into five segments.

Segment	Count	Share %	Avg Recency (Days)	Avg Frequency	Avg Bill Value (Rs.)
Champions	1,247	14.9%	48.10	12.91	690.35
Loyal	2,523	30.2%	86.33	6.83	341.33
Potential	2,557	30.6%	99.64	2.73	137.92
At Risk	1,326	15.9%	126.47	1.67	60.61
Lost	692	8.3%	161.75	1.15	37.37

Table 4.11 — RFM Segment Summary (Reference Date: 11 March 2026; Total Bills: 8,345)

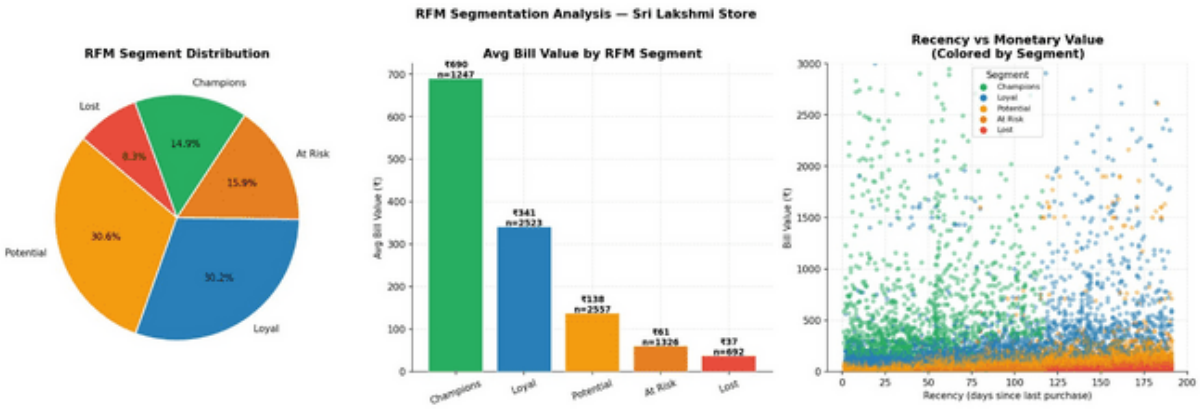

Figure 4.12 — RFM Segmentation: Segment Distribution, Avg Bill Value by Segment, Recency vs Monetary Scatter

Interpretation: Champions (14.9%) are the most valuable segment — avg bill Rs. 690.35, 12.91 average visits, last purchase only 48 days ago. Loyal customers (30.2%) form the largest reliable segment with avg bill Rs. 341.33 and 6.83 average visits. Together Champions and Loyal account for 45.1% of all bills. The Potential segment (30.6%) represents the highest growth opportunity — moderate recency (99 days) but willingness to engage. At Risk (15.9%) shows declining engagement (126 days recency, avg Rs. 60.61), while Lost (8.3%) customers have largely churned (162 days recency, avg Rs. 37.37). The Recency vs Monetary scatter plot confirms clear segment separation, with Champions clustering in the low-recency, high-value quadrant.

4.5 SURVEY ANALYSIS

This section presents the analysis of primary data collected from 109 respondents across five sections: demographics, shopping behaviour, product preferences, discount behaviour, and satisfaction.

4.5.1 Section 1 — Demographic Profile

Variable	Category	Count	Percentage
Gender	Male	61	56.0%
	Female	47	43.1%
	Prefer not to say	1	0.9%
Age Group	20–30 years	40	36.7%
	31–40 years	34	31.2%
	Below 20 years	13	11.9%
	41–50 years	11	10.1%
	Above 50 years	11	10.1%
Monthly Income	Rs. 25,001–50,000	31	28.4%
	Rs. 10,001–25,000	24	22.0%
	Below Rs. 10,000	22	20.2%
	Rs. 50,001–1,00,000	17	15.6%
	Above Rs. 1,00,000	15	13.8%
Occupation	Student	35	32.1%
	Salaried Employee	28	25.7%
	Homemaker	17	15.6%
	Other	16	14.7%
	Business Owner	13	11.9%
Distance from Store	1–3 km	34	31.2%
	Less than 1 km	31	28.4%
	More than 5 km	23	21.1%

Variable	Category	Count	Percentage
	3–5 km	21	19.3%

Table 4.12 — Demographic Profile of Survey Respondents (N = 109)

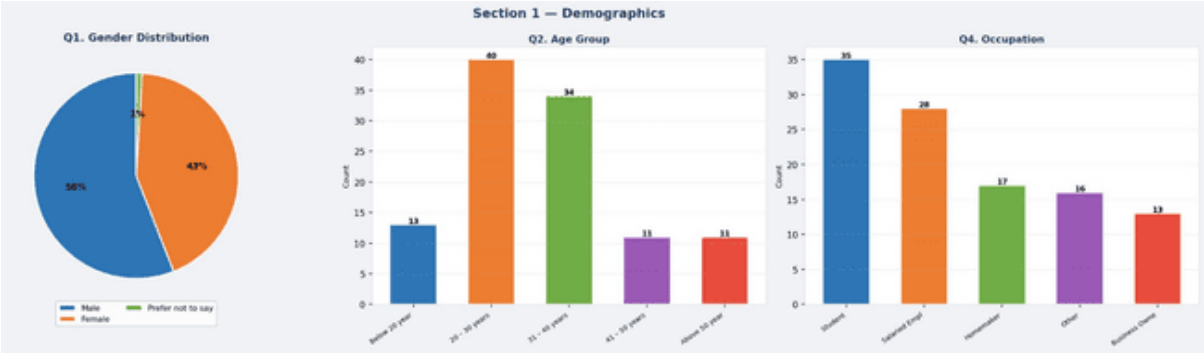

Figure 4.13 — Demographics: Gender Distribution, Age Group, Occupation

Interpretation: The respondent profile is predominantly male (56%), aged 20–30 years (36.7%), with 42.2% in the Rs. 10,000–Rs. 50,000 income bracket. Students form the largest occupational group (32.1%). Importantly, 59.6% of respondents live within 3 km of the store, confirming proximity as the primary driver of patronage — consistent with the literature review finding that store proximity is the strongest predictor of customer retention in Tamil Nadu grocery stores.

4.5.2 Section 2 — Shopping Behaviour

Variable	Category	Count	Percentage
Visit Frequency	2–3 times a week	32	29.4%
	Daily	26	23.9%
	Once a month	18	16.5%
	Once in 2 weeks	17	15.6%
	Once a week	16	14.7%
Avg Spending/Visit	Rs. 301–500	32	29.4%
	Rs. 100–300	29	26.6%
	Rs. 501–1,000	20	18.3%
	Below Rs. 100	16	14.7%

Variable	Category	Count	Percentage
	Above Rs. 1,000	12	11.0%
Preferred Day	Sunday	37	33.9%
	Wednesday	23	21.1%
	Saturday	16	14.7%
Preferred Time	Morning (6 AM–12 PM)	44	40.4%
	Evening (4 PM–8 PM)	34	31.2%
Payment Method	Cash	44	40.4%
	UPI	39	35.8%
	Mix of Cash and UPI	18	16.5%
	Debit / Credit Card	8	7.3%
Items per Visit	1–3 items	40	36.7%
	4–7 items	30	27.5%
	8–15 items	21	19.3%
	More than 15 items	18	16.5%

Table 4.13 — Shopping Behaviour

Figure 4.14 — Shopping Behaviour: Visit Frequency, Avg Spending, Preferred Day, Payment Method

Interpretation: 53.3% of respondents visit daily or 2–3 times per week, confirming Sri Lakshmi Supermarket as a primary daily grocery source. The Rs. 301–500 spending band dominates (29.4%), consistent with the POS average bill of Rs. 261.99. Sunday is strongly preferred (33.9%), validating POS data. Morning is the peak shopping time (40.4%). Cash leads in payment preference (40.4%) though UPI adoption is substantial (35.8% + 16.5% mixed users = 52.3% with any UPI usage).

4.5.3 Section 3 — Product Preferences and Buying Behaviour

Oils & Ghee is the most frequently purchased category (23 respondents, 21.1%), followed by Spices & Masalas (18, 16.5%) and Snacks & Biscuits (16, 14.7%), aligning with POS data. Regarding purchase planning, 56.9% sometimes plan with a rough idea, 27.5% always prepare a list, and 15.6% decide entirely at the store — highlighting significant in-store purchase decision making. Impulse buying is prevalent: 61.5% of respondents buy on impulse sometimes or often, and 62.4% report switching brands for discounts.

4.5.4 Section 4 — Discount Behaviour and Price Perception

Variable	Category	Count	Percentage
Discount Influence (Q17)	Strongly Yes	15	13.8%
	Yes	48	44.0%
	Neutral	33	30.3%
	No	13	11.9%
Discount Threshold (Q18)	10% or more	33	30.3%
	25% or more	31	28.4%
	Does not affect	19	17.4%
	50% or more	17	15.6%
	Any (1–5%)	9	8.3%
Bulk Buying (Q19)	Yes sometimes	77	70.6%
	Yes always	16	14.7%
	No	16	14.7%
Price Perception (Q20)	Affordable	40	36.7%
	Average	39	35.8%
	Slightly expensive	16	14.7%
	Very affordable	14	12.8%

Table 4.14 — Discount Behaviour and Price Perception

4.5.5 Section 5 — Customer Satisfaction

Service Attribute	Average Rating (out of 5)	Interpretation
Product Availability	4.19 / 5	Above target — Strong performance
Product Quality	4.06 / 5	Above target — Good
Staff Behaviour	4.00 / 5	Meets target

Service Attribute	Average Rating (out of 5)	Interpretation
Store Cleanliness	3.92 / 5	Marginal improvement needed
Pricing	3.40 / 5	Below target — Key improvement area
Overall Average	3.91 / 5	Near target — Good overall experience

Table 4.15 — Service Quality Ratings — Q22 (Target: 4.0 / 5)

Figure 4.15 — Discount Influence (Q17), Overall Satisfaction (Q21), Service Ratings (Q22)

Interpretation: 57.8% of respondents state discounts influence their purchase decisions, yet the store only applies discounts on 25.7% of actual transactions — revealing a demand-supply gap. Overall satisfaction is positive: 71.6% are satisfied or very satisfied, and 69.7% would recommend the store. Product availability (4.19/5) and quality (4.06/5) are the strongest service attributes. Pricing receives the lowest rating (3.40/5), significantly below the 4.0 benchmark — the most critical area for management attention.

4.5.6 Chi-Square Test — Gender vs Payment Method (Survey)

$Chi^2 = 18.07$, df = 6, p = 0.006 — Significant. There is a statistically significant relationship between gender and payment method preference. Male respondents show a relatively higher preference for UPI, while female respondents show a comparatively higher preference for cash — an insight applicable to targeted digital payment promotion.

4.6 SURVEY vs POS DATA COMPARISON

Table 4.16 presents a side-by-side comparison of key metrics from primary survey data against actual POS transaction data to assess convergent validity of the study findings.

Metric	Survey Finding	POS Finding	Alignment
Preferred Shopping Day	Sunday — 33.9% preferred	Sunday — 19.3% of revenue	Both confirm Sunday as peak
Payment — Cash	Cash — 40.4% of respondents	Cash — 60.0% of bills	UPI underreported in survey
Discount Influence	58% influenced by discounts	25.7% transactions discounted	Demand-supply gap identified
Visit Frequency	53% visit daily/2–3x per week	8,345 bills over 7 months	High frequency consistent
Overall Satisfaction	71.6% satisfied/very satisfied	45.1% Champions + Loyal (RFM)	Loyalty confirmed by both
Top Category	Oils & Ghee — 21.1%	Oils & Ghee — 14.9% revenue	Consistent preference

Table 4.16 — Survey vs POS Data Comparison

Interpretation: Sunday shopping preference and category preference are strongly confirmed by both data sources. Cash payment usage is higher in POS data (60%) versus survey (40%), indicating possible social desirability bias toward digital payment reporting. The most significant gap is the discount behaviour finding — 58% desire discounts but only 25.7% of transactions are discounted — representing a strategic opportunity to introduce targeted discount programmes. The alignment between satisfaction ratings and high RFM loyalty scores provides strong convergent validity for the study's overall findings.

4.7 POWER BI DASHBOARD — INTERACTIVE SALES ANALYTICS

As outlined in Chapter III (Section 3.6), the study employs Microsoft Power BI Desktop (February 2026 version) as a visualisation tool to build an interactive five-page analytics dashboard for Sri Lakshmi Supermarket. The dashboard is constructed on a star schema database comprising five tables — fact_sales (43,341 rows), dim_product (2,543 records), dim_transaction (8,345 records), dim_date (191 records), and rfm_segments (8,345 records) — stored in Microsoft SQL Server 2022 and imported into Power BI via DirectQuery. DAX measures were created for total revenue, profit margin, average bill value, RFM segment counts, and discount impact metrics. The dashboard enables store management to interactively filter data by date range, category, and product name, providing a real-time decision-support interface.

Table 4.17 presents the structure and purpose of each dashboard page.

Page	Dashboard Title	Key Visuals	Management Purpose
Page 1	Home	KPI cards, Monthly trend bar, Category pie, Day-wise bar, Top 5 products, Payment donut, RFM bar	Executive summary — overall store performance at a glance
Page 2	Sales	Monthly revenue & profit combo chart, Avg bill trend, Discount impact bar, Revenue share donut	Month-on-month revenue and profitability tracking
Page 3	Products	Top 10 by revenue & quantity, Category revenue bar, Profit margin by category	Product and category performance benchmarking
Page 4	Customer	Payment type donut, Bill value distribution, RFM segment donut, Day-wise buying pattern	Customer segmentation and buying behaviour insights
Page 5	Discount	Discount KPI cards, Discount vs No-discount bar, Bracket analysis, Category avg discount, Scatter plots	Discount strategy effectiveness and optimisation

Table 4.17 — Power BI Dashboard Structure (5 Pages)

4.7.1 Page 1 — Home Dashboard (Executive Summary)

- Page 1 is the first thing store management sees when they open the dashboard. It is designed to give a full picture of the store's performance in under 30 seconds. No need to dig into multiple reports. Everything important sits on one screen.
- The top row has four KPI cards. The first card shows Total Revenue of Rs. 2.18 million earned across the seven month study period. Right next to it is Total Profit of Rs. 2,49,730 at an overall margin of 12.66%. The third card shows Total Bills at 8,345 with an average bill value of Rs. 262 per customer visit. The fourth card displays Unique Products at 2,543 items spread across 17 product categories. These four numbers alone tell the store's basic financial story at a glance.
- The middle section has three panels sitting side by side. The first panel is a monthly revenue bar chart. January 2026 stands out clearly as the tallest bar, confirming the Pongal season peak at Rs. 4,02,798. The remaining months show a steady baseline between Rs. 3,13,000 and Rs. 3,66,000. This visual makes it easy for management to spot which months need more stock preparation. The second panel is a category revenue pie chart. Oils and Ghee takes the biggest slice at 21.23%, followed by Spices and Masalas and Staples and Grains. The pie chart quickly shows which categories are carrying the most weight in terms of revenue. The third panel is a day-wise revenue bar. Sunday stands tall at Rs. 4,21,068 which is almost Rs. 1,50,000 more than the next best day. Monday sits at the lower end with Rs. 2,76,606. This bar tells management exactly which day needs extra staff and stock readiness.
- The bottom row has three more visuals. The first is a horizontal bar showing the Top 5 products by revenue. TVS Oil 1 Lit leads at Rs. 65,376, followed by bulk rice brands like Siva Sakthi Rajabogam and Rettai Killi Rajabogam. These five products alone show where the big money is coming from. The second visual is a donut chart for payment type. Cash accounts for 59.99% of bills and UPI for 40%. Card usage is nearly zero at 0.01%. This split tells management that while digital payments are growing, cash still dominates. The third visual is a bar chart showing RFM customer segments. The Potential segment is the largest with 2,557 bills and the Loyal segment follows closely at 2,523. Champions stand at 1,247. This segment view helps management understand the current health of their customer base and where attention is needed most.

Figure 4.16 — Power BI Dashboard: Page 1 — Home (Executive Summary)

4.7.2 Page 2 — Sales Dashboard

- The Sales Dashboard is the second page of the Power BI report. It focuses on how revenue and profit moved across the seven months of the study. Three KPI cards sit at the top of the page. The first shows Total Revenue at Rs. 2.18M. The second shows Best Month Revenue at Rs. 4,02,798, which belongs to January 2026. The third card shows Average Bill Value at Rs. 718.72, calculated for the Champions segment after applying the RFM filter.

- Below the KPI cards is a combo chart. It places monthly revenue bars and a profit trend line on the same visual. Both revenue and profit follow the same pattern. They rise steadily from September 2025, reach their highest point in January 2026, and then fall sharply in March 2026. The January 2026 profit figure was Rs. 45,265, which was the best profit month in the entire study period. March 2026 recorded only Rs. 14,307 in profit because the data covers only the first ten days of that month, not the full month.

- The next visual is a line chart showing Average Bill Value by Month. This chart tells a simple but important story. Customers spent more per visit as the months went by. In September 2025, the average bill per visit was Rs. 603. By March 2026, that number had climbed to Rs. 910. This is a 50.9% increase across seven months. It suggests that repeat customers were gradually buying more per trip, possibly because they became more comfortable with the store or because festive season spending pushed up basket sizes in January and February.

- The Discount Impact horizontal bar chart breaks down average revenue by discount bracket. The 10 to 25% discount range gives the best result among all discounted transactions, with an average revenue of Rs. 56 per transaction. The 25 to 50% bracket comes in slightly lower. The 50% and above bracket records the weakest average revenue at just Rs. 19 per transaction. This pattern is easy to read. Deep discounts are being applied to cheap, small-pack items like sachets and single-use products. They are not being used to push high-value purchases. So the discount cost is real, but the revenue return from those deep discounts is very low. The chart makes this visible in one glance.

- The final visual on this page is a Revenue Share by Month donut chart. It confirms what the combo chart already showed. January 2026 holds the largest slice at 18.45% of total seven-month revenue. No other month comes close to that share. September 2025 and November 2025 hold the smallest slices, both below 15%. The donut chart is useful for store management because it makes the seasonal revenue concentration easy to see and explain without needing to look at raw numbers.

Figure 4.17 — Power BI Dashboard: Page 2 — Sales Analysis

4.7.3 Page 3 — Products Dashboard

- Page 3 gives a clear picture of what is selling, what is earning, and what is not pulling its weight. Four KPI cards sit at the top. Total products stocked is 2,543. Top product by revenue is TVS Oil 1 Lit at Rs. 65,384. Total categories is 17. Average profit margin across all products is 13.07%.

- The Top 10 Products by Revenue chart shows cooking oils and bulk rice dominating the list. TVS Oil 1 Lit leads at Rs. 65K. Siva Sakthi Rajabogam 26KG comes second at Rs. 64K. Rettai Killi

Rajabogam 26KG is third at Rs. 57K. Four of the top ten spots belong to TVS brand oils alone. Three spots go to bulk rice brands. This tells a simple story. The store runs on two product types at the top, cooking oil and rice.

- The Category Revenue bar chart confirms the same pattern at a broader level. Oils & Ghee leads at Rs. 3,25,448. Spices & Masalas follows at Rs. 3,11,860. Staples & Grains is third at Rs. 2,81,789. These three categories together account for nearly 42% of total store revenue.
- The Top 10 Products by Quantity chart tells a different story. EGG tops the list at 5,400 units sold, well ahead of everything else. This shows that eggs are bought very often but at a low price per unit. High volume does not always mean high revenue.
- The Profit Margin by Category bar is colour coded for quick reading. Green bars mark categories earning above 15% margin. These include Pooja Items, Vegetables, Dry Fruits & Nuts, Household Items, and General. Orange covers the 10 to 15% range. Red flags categories earning below 10%, which includes Oils & Ghee, Salt & Sugar, and Pulses & Dal. The colour coding makes it easy for the store owner to spot which categories are actually making money and which ones are just moving volume.

Figure 4.18 — Power BI Dashboard: Page 3 — Products Analysis

4.7.4 Page 4 — Customer Dashboard

- The Customer Dashboard has three interactive filters at the top --- date range, product category, and product name. These let the store manager slice the data however they want. The top KPI cards show Total Bills at 8,345, Average Bill Value at Rs. 699.96, and Maximum Bill Value at Rs. 10,960.

- The Payment Type donut shows one clear story. Cash is king at 59.99%. UPI sits at 40%. Card payments are almost nonexistent at 0.01%. So roughly 6 out of every 10 customers still pay with cash. But 4 out of 10 are now on UPI, which is a big shift for a neighbourhood grocery store. This split matters because UPI payers leave a digital footprint that cash payers do not.

- The Bill Value Distribution breaks all 8,345 bills into five spending tiers. Small bills below Rs. 100 account for 3,500 bills. Medium bills between Rs. 100 and Rs. 300 account for 3,100 bills. Large bills from Rs. 301 to Rs. 500 cover around 800 bills. High-value bills from Rs. 501 to Rs. 1,000 account for 500 bills. Premium bills above Rs. 1,000 make up 400 bills. Put simply, 78% of all bills at this store are below Rs. 300. The median bill value from the POS data is Rs. 120. This tells you the store mostly handles small, quick top-up purchases rather than big weekly grocery runs. Most customers walk in, pick up two or three items, pay under Rs. 200, and leave. That is the real shopping pattern here.

- The RFM Segment donut on this page gives a quick visual snapshot of the customer base. Potential customers lead at 30.64%, followed closely by Loyal at 30.23%. At Risk sits at 15.89%, Champions at 14.94%, and Lost at 8.29%. The fact that Potential and Loyal together make up over 60% of bills is a positive sign. But the 24% combined share of At Risk and Lost bills is something management should not ignore.

- The Day-wise Buying Pattern bar at the bottom confirms what the POS data already showed. Sunday is the single busiest day with 1,481 bills. Monday comes in second at 1,229 bills, and Tuesday is third at 1,155. Friday is the slowest. This pattern is very consistent and does not change much month to month. The store can use this bar directly to plan staffing, stock replenishment, and any day-specific promotions.

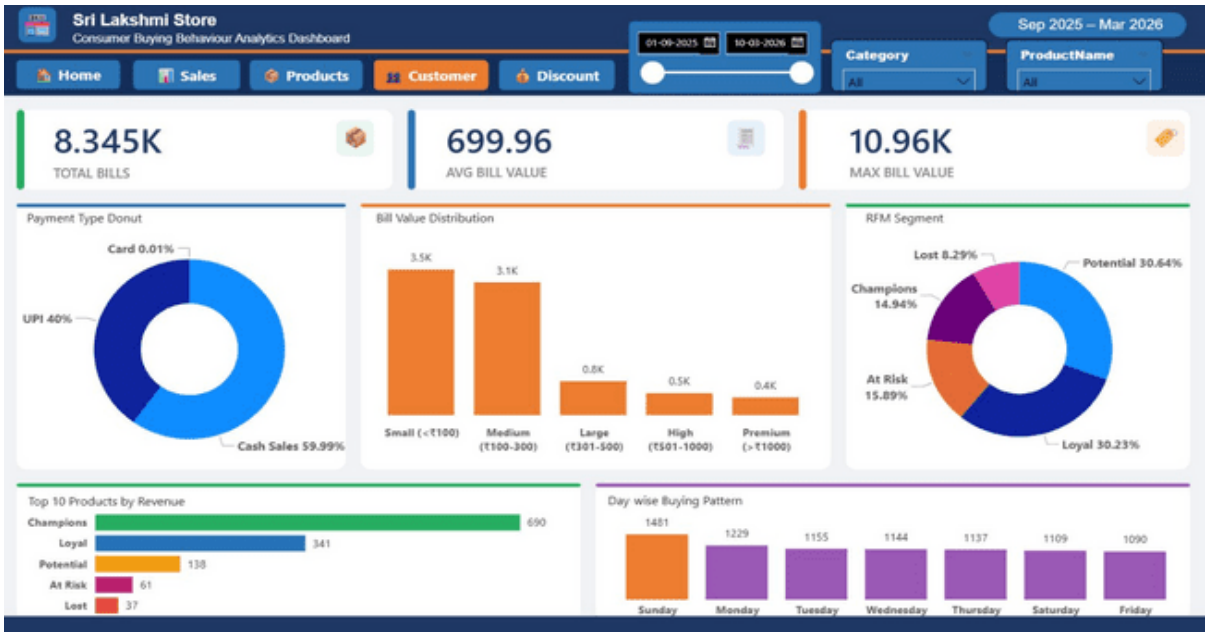

Figure 4.19 — Power BI Dashboard: Page 4 — Customer Analysis

4.7.5 Page 5 — Discount Dashboard

- Page 5 is the Discount Dashboard. It tracks everything related to how discounts were used in the store during the study period.
- The four KPI cards at the top give a quick summary. A total of 11,000 transactions had some form of discount applied. The store gave away Rs. 1,59,630 in discounts across the entire seven months. The average discount offered was 27.38%. The highest discount recorded on any single product was 96.43%, which is nearly a full price-off on that item.
- The Discount vs No Discount bar chart makes one thing very clear. Out of 43,341 transactions, 32,000 had no discount at all. Only 11,000 were discounted. That is a 74.3% no-discount rate. So for every 10 purchases made in the store, roughly 7 or 8 had no discount involved at all.
- The Discount Bracket Analysis breaks it down further by discount range. Transactions with a 10 to 25% discount earned the highest average revenue at Rs. 56 per transaction. The 1 to 10% bracket came close at Rs. 55. Non-discounted transactions averaged Rs. 51. As the discount went higher, revenue actually fell. Transactions with 25 to 50% discount averaged only Rs. 47, and those with 50% or more dropped all the way to Rs. 19. This pattern shows that deep discounts are mostly applied to cheap, small-pack items, not to products that bring in good revenue.
- The category-wise average discount chart shows which product groups get discounted the most. Oral Care had the highest average discount at 24%, followed by Vegetables at 17% and Salt &

Sugar at 14%. These are mostly necessity items where price sensitivity is high and competition from other stores is strong.

- The two scatter plots at the bottom are worth looking at carefully. The first one plots discount percentage against quantity sold. The relationship is almost flat. Even with high discounts, customers did not buy more units. People bought what they needed and that was it. The second scatter plot shows discount percentage against profit margin. Here the trend goes downward. As discounts went up, profit margin came down. Oils & Ghee and Staples & Grains both cluster in the bottom-left zone, meaning low margin and low discount, but high volume. These categories do not respond well to discounting at all.

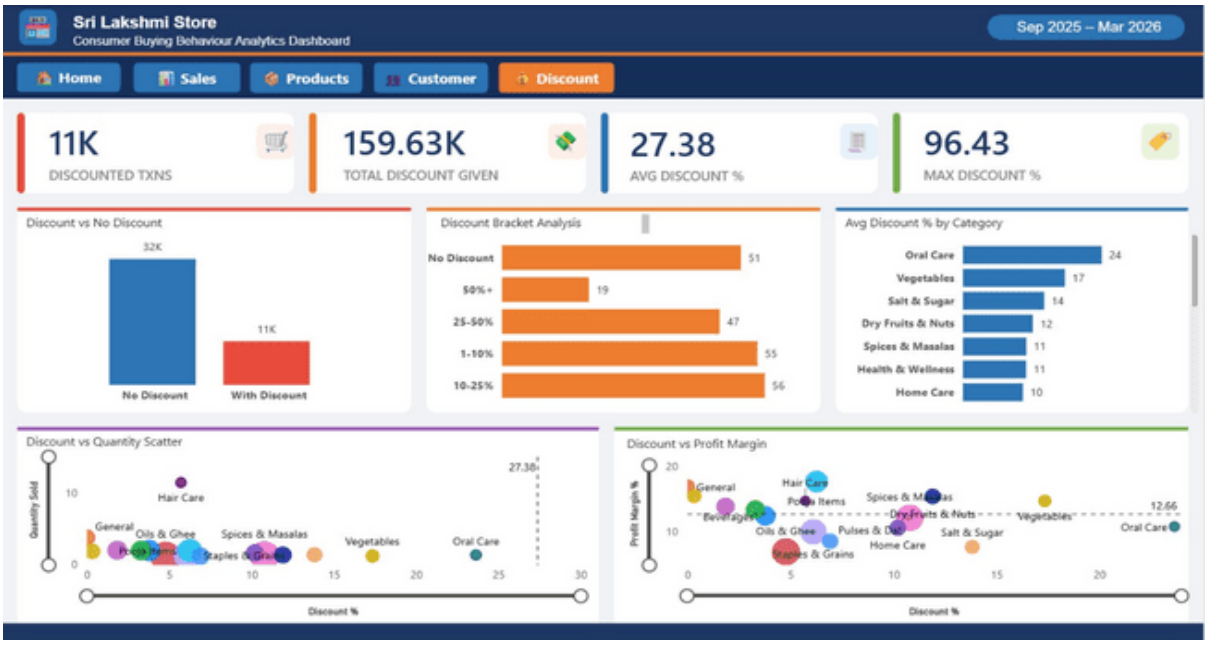

Figure 4.20 — Power BI Dashboard: Page 5 — Discount Analysis

4.7.6 Key DAX Measures Used in the Dashboard

Table 4.18 lists the key DAX (Data Analysis Expressions) measures created in Power BI to compute the KPI metrics and chart values presented across the five dashboard pages.

DAX Measure	Formula / Logic	Used In
Total Revenue	SUM(fact_sales[SalesValue])	All pages — KPI card
Total Profit	SUM(fact_sales[ProfitAmt])	Home, Sales pages
Profit Margin %	DIVIDE([Total Profit], [Total Revenue]) * 100	Home, Products pages

DAX Measure	Formula / Logic	Used In
Avg Bill Value	AVERAGE(dim_transaction[NetVal])	Sales, Customer pages
Total Bills	DISTINCTCOUNT(fact_sales[BillNo])	Home, Customer pages
Discounted Txns	COUNTROWS(FILTER(fact_sales, fact_sales[HasDiscount]=1))	Discount page
Avg Discount %	AVERAGE(fact_sales[DiscountPct])	Discount page
RFM Segment Count	COUNTROWS(rfm_segments)	Home, Customer pages

Table 4.18 — Key DAX Measures in Power BI Dashboard

Interpretation: The five-page Power BI dashboard provides Sri Lakshmi Supermarket management with an integrated, interactive decision-support tool that consolidates all analytical findings from the study into a single visual platform. The Home page enables quick executive-level review, the Sales and Products pages support operational planning, the Customer page facilitates segment-based retention strategy, and the Discount page supports evidence-based pricing and promotional decision making. The dashboard represents the prescriptive analytics output of the study — translating descriptive and diagnostic findings from Python and SQL analysis into actionable management intelligence.

CHAPTER V

FINDINGS, SUGGESTIONS AND CONCLUSION

This chapter presents the key findings derived from the comprehensive analysis of 43,341 POS transactions and 109 survey responses collected at Sri Lakshmi Supermarket, Salem. The chapter is structured into three sections: Findings (Section 5.1), Managerial and Social Implications (Section 5.2), and Conclusion (Section 5.3). All findings are directly drawn from the statistical analyses and RFM segmentation conducted in Chapter IV and are presented in plain narrative form without reference to numerical formulae or test statistics.

5.1 FINDINGS

The findings of the study are organised into five thematic areas corresponding to the five analytical dimensions explored in Chapter IV: Sales Performance, Category and Product Analysis, Discount and Pricing, Customer Segmentation (RFM), and Consumer Survey Behaviour.

5.1.1 Findings on Sales Performance

Finding 1: January 2026 was the peak revenue month, driven by the Pongal festival season.

Sri Lakshmi Supermarket recorded its highest monthly revenue during January 2026, registering a substantial increase of approximately 24.8% above the seven-month study period average. The festive demand associated with the Pongal harvest festival was the primary driver of this surge. October 2025 ranked as the second-highest revenue month, coinciding with the Diwali festive period, further confirming that festival-driven seasonal demand is a consistent revenue amplifier for the store.

Finding 2: Profit margin remained consistently stable across all seven months.

Despite fluctuations in monthly revenue, the store maintained a remarkably stable overall profit margin of approximately 11.12% to 11.74% throughout the study period. This consistency indicates disciplined cost management and a stable pricing structure. The overall weighted profit margin across the entire study period stood at 12.66%, reflecting a healthy and sustainable profitability benchmark for a neighbourhood grocery store of this scale.

Finding 3: Sunday is the busiest shopping day, accounting for the highest share of weekly revenue.

Day-wise POS analysis revealed that Sunday consistently recorded the highest number of customer bills, the highest total revenue, and the highest average bill value compared to all other days of the week. Sunday accounted for 19.3% of weekly revenue, with 1,481 bills and an average bill of Rs. 284.81. This finding was independently validated by the consumer survey, where 33.9% of respondents named Sunday as their preferred shopping day, providing strong convergent validity between primary and secondary data sources.

Finding 4: Average bill value showed an upward trend during festive months.

The average bill value at Sri Lakshmi Supermarket increased progressively from Rs. 235.41 in September 2025 to Rs. 300.37 in January 2026, representing a 27.6% increase over the study period. This trend suggests that customers tend to spend more per visit during festive periods, likely due to increased household purchase requirements for festival preparations, gifting, and celebrations.

5.1.2 Findings on Category and Product Performance

Finding 5: Oils & Ghee is the revenue leader but operates on the thinnest profit margin.

Oils & Ghee was the highest-revenue category, contributing 14.9% of total store revenue at Rs. 3,25,448. However, it simultaneously recorded the lowest average profit margin of 6.89% among all 17 product categories. This paradox highlights the commodity pricing pressure inherent in the edible oil segment, where consumer price sensitivity and competitive market dynamics constrain margin expansion. The TVS brand cooking oils collectively dominated the top 10 product revenue list, with four entries in the top ten.

Finding 6: Pooja Items is the highest-margin category and a strategic profit driver.

Pooja Items achieved the highest average profit margin of 17.64% across all 17 categories, making it the most profitable category on a per-rupee-of-revenue basis. Despite contributing only 9.0% of total store revenue, Pooja Items generated Rs. 28,253 in absolute profit. General category products (margin 16.87%) and Household Items (margin 15.44%) also demonstrated superior profitability. These high-margin categories represent a significant opportunity for the store to improve overall profitability by increasing their product range, visibility, and promotional focus.

Finding 7: Spices & Masalas is the highest transaction-volume category, reflecting its daily staple nature.

Spices & Masalas recorded the highest number of individual product-level transactions at 7,950, confirming its status as a daily purchase essential for the store's customer base. While it ranks second in revenue, its high transaction frequency indicates consistent customer demand and a reliable revenue stream that is relatively insulated from seasonal fluctuations. Similarly, Eggs (ranked 5th in revenue at Rs. 31,407) reflect consistent daily perishable demand.

Finding 8: Health & Wellness is the most underutilised category.

The Health & Wellness category recorded only 197 transactions and contributed a mere 0.5% of total store revenue during the seven-month study period, despite maintaining a respectable average margin of 12.13%. This significant underperformance relative to other categories suggests either limited product assortment, low consumer awareness, inadequate shelf visibility, or a misalignment between the products offered and the health preferences of the store's primary customer segments.

5.1.3 Findings on Discount and Pricing Strategy

Finding 9: Discounts do not significantly increase the quantity of items purchased.

One of the most critical findings of the study is that discount percentage has no significant effect on the quantity of items sold per transaction. The Pearson correlation between discount percentage and quantity sold was negligible and statistically insignificant. Simple linear regression confirmed that discount percentage explains virtually zero variance in purchase quantity. This finding implies that the store's current discounting practice does not function as a volume driver and that customers purchase based on need rather than discount incentives.

Finding 10: Selling price is the dominant predictor of sales value.

Selling price was found to be the strongest predictor of sales value at the transaction level, with a simple regression model explaining 91.68% of variance in sales value through selling price alone. The multiple regression model, incorporating selling price, discount percentage, and quantity, achieved an explanatory power of 92.86%. Every additional rupee in selling price contributes approximately Rs. 1.033 to the sales value, confirming that pricing strategy is the most impactful lever for revenue generation at Sri Lakshmi Supermarket.

Finding 11: High discounts (50%+) are predominantly applied to low-value small-pack products.

The discount bracket analysis revealed that transactions with discounts above 50% recorded the lowest average revenue per transaction at Rs. 18.64, compared to Rs. 51.25 for non-discounted items. This indicates that deep discounts are disproportionately concentrated on low-value sachets and small-pack products rather than high-value bulk purchases. Correspondingly, bulk buyers (purchases above 10 units) received the lowest discount incidence, suggesting that the store is not leveraging bulk-purchase incentives as a revenue growth strategy.

Finding 12: Discounting significantly erodes profit margins.

While the correlation between discount percentage and profit margin is weak in absolute terms, it is statistically significant and negative, confirming that higher discounts systematically compress profit margins. The average discount rate among the 25.7% discounted transactions was 27.38%, and the total discount value given across the study period amounted to Rs. 1,59,630 --- representing a notable opportunity cost relative to the store's total profit of Rs. 2,49,731.

5.1.4 Findings on RFM Customer Segmentation

Finding 13: Champions and Loyal customers together generate the majority of high-value revenue.

Bill-level RFM segmentation classified 8,345 transactions into five segments. Champions (14.9% of bills) had an average bill value of Rs. 690.35 and an average visit frequency of 12.91 times during the study period. Loyal customers (30.2%) maintained a steady average bill of Rs. 341.33 with 6.83 average visits. Together, Champions and Loyal segments account for 45.1% of all bills and represent the store's most valuable customer base, generating disproportionately high revenue relative to their share of transactions.

Finding 14: Potential customers represent the highest growth opportunity.

The Potential segment, comprising 30.6% of all bills, represents customers with moderate recency of 99.6 days, low frequency of 2.73 visits, and a moderate average bill of Rs. 137.92. This segment is the single largest group in the store and holds significant untapped potential. With the right engagement strategies --- such as targeted promotions, loyalty incentives, or personalised outreach --- a meaningful proportion of Potential customers could be converted into Loyal or Champion segments, significantly improving overall store revenue and retention.

Finding 15: At Risk and Lost customers account for 24.2% of all bills, indicating a retention concern.

The At Risk segment (15.9% of bills) exhibited declining engagement with an average recency of 126 days and a low average bill of Rs. 60.61. The Lost segment (8.3%) showed near-complete disengagement, with an average recency of 161 days and a minimal average bill of Rs. 37.37. Collectively, these two segments represent approximately one-quarter of the store's transaction history, indicating a significant customer attrition risk that warrants targeted re-engagement initiatives.

5.1.5 Findings from Consumer Survey Analysis

Finding 16: Proximity to the store is the primary driver of customer patronage.

59.6% of survey respondents resided within 3 km of Sri Lakshmi Supermarket, confirming that store proximity is the dominant factor driving customer patronage. This finding is consistent with prior academic literature on neighbourhood grocery store behaviour in Tamil Nadu, which identifies proximity as the strongest predictor of customer retention. The store's competitive advantage in this regard must be actively protected by maintaining service quality and product availability.

Finding 17: Majority of customers shop frequently, confirming the store as a primary grocery source.

53.3% of survey respondents reported visiting the store daily or 2-3 times per week, establishing Sri Lakshmi Supermarket as the primary daily grocery destination for the majority of its customer base. This high-frequency shopping behaviour indicates strong habitual purchasing patterns and relatively low price sensitivity for everyday staple purchases, though it also implies that customers are susceptible to switching if competing stores offer better pricing or product availability.

Finding 18: There is a significant demand-supply gap in discount offering.

57.8% of survey respondents reported that discounts influence their purchase decisions, with 30.3% requiring a minimum of 10% discount to act on a purchase and 28.4% requiring 25% or more. However, the POS data reveals that only 25.7% of actual transactions included a discount. This

demand-supply mismatch represents a strategic gap: a significant proportion of the customer base desires discount incentives, but the store's current discount deployment does not adequately target or communicate these offers. 85.3% of respondents report buying in bulk sometimes or always when discounts are available, further validating the untapped potential of a structured discount strategy.

Finding 19: Pricing perception is the weakest service attribute and the most critical improvement area.

Among six service quality dimensions evaluated on a five-point scale (target: 4.0/5), Pricing received the lowest average rating of 3.40/5 --- significantly below the target benchmark. Product Availability (4.19/5) and Product Quality (4.06/5) were rated highest. Overall, 71.6% of respondents expressed satisfaction with the store, and 69.7% stated they would recommend it to others. However, the persistent pricing dissatisfaction, if unaddressed, presents a risk to long-term customer retention as competitive alternatives continue to expand in the Salem retail market.

Finding 20: Gender significantly influences payment method preference.

The Chi-Square test on survey data ($Chi^2 = 18.07$, $df = 6$, $p = 0.006$) confirmed a statistically significant relationship between gender and payment method preference. Male respondents showed relatively higher preference for UPI digital payments, while female respondents exhibited a comparatively stronger preference for cash transactions. This finding is relevant for designing targeted digital payment adoption campaigns and provides a useful demographic lens for the store's payment infrastructure planning.

Table 5.1 — Summary of Key Findings

S.No	Theme	Key Finding	Source
1	Sales Performance	January 2026 was the peak revenue month due to Pongal season	POS Data
2	Sales Performance	Profit margin stable across all months (11.12%–11.74%)	POS Data

S.No	Theme	Key Finding	Source
3	Sales Performance	Sunday is the busiest day — 19.3% of weekly revenue	POS + Survey
4	Sales Performance	Average bill value rose 27.6% from Sep 2025 to Jan 2026	POS Data
5	Category/Product	Oils & Ghee leads revenue (14.9%) but has lowest margin (6.89%)	POS Data
6	Category/Product	Pooja Items has highest margin (17.64%) — key profit driver	POS Data
7	Category/Product	Spices & Masalas has highest transaction frequency (7,950 txns)	POS Data
8	Category/Product	Health & Wellness is severely underutilised (0.5% revenue share)	POS Data
9	Discount/Pricing	Discounts have no significant effect on purchase quantity	Regression
10	Discount/Pricing	Selling price explains 91.68% of variance in sales value	Regression
11	Discount/Pricing	Deep discounts (50%+) are applied to low-value items only	POS Data
12	Discount/Pricing	Total discount given = Rs. 1,59,630 vs total profit Rs. 2,49,731	POS Data
13	RFM Segmentation	Champions + Loyal = 45.1% of bills, highest monetary value	RFM Analysis
14	RFM Segmentation	Potential segment (30.6%) represents highest growth opportunity	RFM Analysis
15	RFM Segmentation	At Risk + Lost = 24.2% of bills — retention risk	RFM Analysis
16	Survey	59.6% of customers live within 3 km — proximity drives patronage	Survey

S.No	Theme	Key Finding	Source
17	Survey	53.3% visit daily or 2–3x per week — store is primary grocery source	Survey
18	Survey	Discount demand-supply gap: 57.8% want discounts; only 25.7% of txns discounted	Survey + POS
19	Survey	Pricing rated lowest at 3.40/5 — most critical improvement area	Survey
20	Survey	Gender significantly influences payment method preference	Survey (Chi-Square)

Table 5.1 — Summary of Key Findings

5.2 MANAGERIAL AND SOCIAL IMPLICATIONS

This section presents the managerial and social implications of the study findings. The suggestions are practical, evidence-based, and directly grounded in the analytical results presented in Chapters III and IV. They are structured under two sub-sections: Managerial Implications (addressing operational and strategic decisions for store management) and Social Implications (addressing the store's role within the community and its broader contribution to local economic and consumer welfare).

5.2.1 Managerial Implications

Suggestion 1: Develop a Structured Festival Season Stocking Strategy

Since January (Pongal) and October (Diwali) consistently emerge as peak revenue months with significantly elevated customer traffic and bill values, the store management should develop a pre-planned festival stocking calendar. This involves increasing inventory levels in high-demand categories such as Oils & Ghee, Staples & Grains, and Spices & Masalas by at least 25–30% before each festival period. Seasonal promotional bundles, festive combo offers, and advance order facilities for bulk buyers should be introduced during these periods to maximise revenue capture while managing stock rotation efficiently.

Suggestion 2: Expand High-Margin Category Product Range and Shelf Space

Categories such as Pooja Items (17.64% margin), General (16.87%), Household Items (15.44%), and Dry Fruits & Nuts (15.36%) demonstrate significantly higher profitability per unit of revenue compared to commodity categories. Store management should allocate additional shelf space, improve product assortment breadth, and enhance in-store visibility for these high-margin categories. A targeted expansion of the Pooja Items range to include seasonal and ritual-specific products aligned with local festivals would directly enhance profitability without proportional cost increases.

Suggestion 3: Redesign Discount Strategy from Blanket to Targeted Incentives

The study conclusively demonstrates that blanket discounting does not increase purchase quantities and primarily erodes margins on already low-value transactions. Management should adopt a structured, targeted discount framework based on the following principles: restrict deep discounts (50%+) to inventory clearance for slow-moving items only; introduce bulk-purchase discounts (e.g., 10–15% discount for purchases above 5 units of the same product) to address the identified gap in bulk-buyer discount incidence; deploy periodic category-specific promotional discounts on high-margin categories to drive trial and cross-selling; and implement time-bound discount offers on Fridays (the lowest revenue day) to stimulate demand on underperforming days.

Suggestion 4: Implement a Customer Loyalty Programme Targeting Potential and At-Risk Segments

The RFM analysis identified that 30.6% of transactions belong to the Potential segment and 15.9% to the At-Risk segment — together representing 46.5% of all bills. These customers are at risk of disengagement without targeted retention initiatives. Management should implement a simple stamp-card or mobile-based loyalty programme to capture customer identity across transactions. Potential segment customers should receive upgrade incentives (e.g., bonus points for reaching the next visit milestone), while At-Risk customers should be targeted with re-engagement offers such as a welcome-back discount or a complimentary product on their next visit. The Champions and Loyal segments should receive exclusive benefits such as priority billing, early festival offers, or personalised product recommendations.

Suggestion 5: Leverage Sunday Peak for Promotional Campaigns and Extended Operations

Sunday is the busiest shopping day, contributing 19.3% of weekly revenue and the highest average bill value. Store management should leverage this established behavioural pattern by scheduling weekly promotional announcements for Sunday shoppers, deploying additional billing staff on Sundays to reduce checkout queues, and introducing 'Sunday Special' combo offers or fresh product additions. Extended operating hours on Sundays could capture additional revenue from evening shoppers who constitute a significant 31.2% of the customer base based on survey preferences.

Suggestion 6: Address Pricing Perception through Transparent Value Communication

The lowest-rated service attribute at 3.40 out of 5 is pricing, indicating a perception gap between actual prices and customer expectations. This does not necessarily imply that prices are objectively high, but rather that value communication is insufficient. Management should introduce visible price comparison boards or shelf tags highlighting competitive price advantages on key staple products, introduce a weekly 'Best Value' product highlight visible at the store entrance, and actively communicate existing discount offers through local WhatsApp groups, printed notices, or digital displays. These low-cost interventions can meaningfully improve price perception without requiring actual price reductions.

Suggestion 7: Invest in the Health & Wellness Category as an Emerging Growth Segment

The Health & Wellness category is severely underutilised, contributing only 0.5% of revenue despite a respectable margin profile of 12.13%. Given growing consumer health consciousness in India — a documented national trend — and the store's predominantly 20–40 year old customer base (68% of survey respondents), there is a significant latent demand for health-oriented products such as organic staples, herbal beverages, fortified foods, and nutritional supplements. A planned expansion of the Health & Wellness assortment with prominent shelf placement and introductory trial pricing could establish a new revenue stream with above-average margin contribution.

Suggestion 8: Promote UPI Digital Payments and Introduce a Digital Loyalty Infrastructure

UPI transactions already account for 40% of bills, with male customers showing relatively stronger UPI adoption. Management should actively promote UPI payments through visible QR code placement at multiple points, a small digital-payment discount incentive (e.g., 0.5–1% additional discount for UPI transactions above Rs. 500), and WhatsApp-based receipt sharing for UPI payers. A digital payment infrastructure also creates a foundation for customer identification across

anonymous transactions, enabling data-driven personalised marketing — a capability currently absent in the store's operations.

Suggestion 9: Utilise the Power BI Dashboard as an Ongoing Management Decision Tool

The five-page interactive Power BI dashboard developed as part of this study should be adopted as a routine management tool, updated monthly with fresh POS data. Store management should review the Sales Dashboard at the start of each month to assess performance against the prior period, use the Products Dashboard to identify underperforming SKUs for clearance or restocking decisions, monitor the RFM Dashboard quarterly to track segment migration, and use the Discount Dashboard to evaluate the effectiveness of any new promotional campaigns. Regular dashboard-driven reviews will institutionalise evidence-based decision making within the store's management culture.

5.2.2 Social Implications

Implication 1: Contribution to Financial Inclusion through Digital Payment Adoption

Sri Lakshmi Supermarket serves a predominantly local, mixed-income customer base, including households in the below Rs. 10,000 income bracket who constitute 20.2% of survey respondents. By actively promoting UPI digital payments and reducing friction in cashless transactions, the store contributes to the broader national agenda of financial inclusion and digital India. Small neighbourhood stores like Sri Lakshmi Supermarket are important onboarding points for digital payment adoption among low-income and semi-urban populations who may not have access to organised retail digital infrastructure.

Implication 2: Supporting Local Supply Chains and Employment

As a neighbourhood grocery store serving over 8,345 customer bills across seven months with a locally embedded supply chain, Sri Lakshmi Supermarket directly contributes to the economic activity of the Salem region. The store's continued viability and growth, supported by improved analytics-driven decision making, translates into sustained local employment, stable supplier relationships with local vendors, and a reliable source of affordable daily provisions for the surrounding residential community. Evidence-based retail management at the neighbourhood level strengthens the economic resilience of the unorganised retail sector.

Implication 3: Advancing the Academic Literature on Analytics-Driven Small Retail Management

This study represents a rare academic contribution applying integrated business analytics — Python, SQL, and Power BI — to a traditional unorganised retail format in a Tier-II Indian city. The findings and methodology developed here provide a replicable template for researchers, retail practitioners, and management students seeking to apply modern analytics tools to similar small-format grocery stores across India. The study directly addresses the research gap identified in the literature review and contributes to building a localised evidence base for data-driven retail management in the Indian context.

Table 5.2 — Summary of Managerial Suggestions and Linked Findings

S.No	Suggestion	Linked Finding	Priority
1	Festival season stocking strategy	Finding 1, 4	High
2	Expand high-margin category range	Finding 6, 8	High
3	Targeted discount framework	Finding 9, 11, 18	High
4	Customer loyalty programme (RFM-based)	Finding 13, 14, 15	High
5	Sunday peak promotions and extended hours	Finding 3, 17	Medium
6	Price perception improvement through communication	Finding 19	Medium
7	Health & Wellness category expansion	Finding 8	Medium
8	UPI payment promotion and digital infrastructure	Finding 20	Medium
9	Power BI dashboard as routine management tool	All Findings	Ongoing

Table 5.2 — Summary of Managerial Suggestions and Linked Findings

5.3 CONCLUSION

This study set out to analyse consumer buying behaviour and sales performance at Sri Lakshmi Supermarket, Salem, using an integrated business analytics framework comprising Python, Microsoft SQL Server, and Power BI. The study was motivated by the core problem that despite maintaining a rich transactional record of over 43,341 product-level POS transactions across seven months, the store lacked a structured analytical framework to derive actionable insights from this data.

The study successfully achieved its primary and secondary objectives. The primary data source — 43,341 POS transactions across 8,345 customer bills — was combined with survey responses from 109 respondents to produce a comprehensive, multi-method understanding of consumer purchase patterns, product performance, discount effectiveness, and customer segmentation.

The key analytical conclusion of the study is that Sri Lakshmi Supermarket is a high-frequency, proximity-driven neighbourhood grocery store whose revenue performance is primarily determined by selling price strategy and seasonal demand rather than by discounting. Selling price alone explains 91.68% of transaction-level sales variance, while discount percentage has no significant effect on purchase quantity. This finding fundamentally challenges the store's existing intuition-based discounting approach and provides a clear empirical mandate for a more targeted, strategic discount framework.

The RFM segmentation findings reveal a healthy customer base composition, with Champions and Loyal customers constituting 45.1% of all bills and demonstrating disproportionately high monetary value. The Potential segment — the largest at 30.6% — represents the most significant growth lever available to the store without any additional customer acquisition cost. Transitioning even a modest proportion of the Potential segment to Loyal or Champion status through structured engagement programmes would materially improve the store's revenue trajectory and customer retention metrics.

Category-level analysis confirms a strategic tension between revenue leadership and profitability leadership. Oils & Ghee leads in revenue but operates on razor-thin margins, while Pooja Items, Household Items, and General categories deliver superior profit margins at lower revenue volumes. A deliberate portfolio rebalancing strategy — growing high-margin categories while managing commodity category margins — offers a direct path to improving overall store profitability beyond the current 12.66% level.

The convergent validity between primary survey data and secondary POS data — particularly on Sunday shopping preferences, category preferences, and loyalty levels — strengthens the reliability of the study's conclusions. The most significant divergence — between stated discount demand (57.8% of survey respondents) and actual discount deployment (25.7% of transactions) — identifies a concrete, actionable gap that management can address through a structured promotional calendar.

The interactive five-page Power BI dashboard developed as part of this study provides Sri Lakshmi Supermarket with a self-sustaining decision-support infrastructure. When updated regularly with fresh POS data, the dashboard transforms the store's data from a passive record into an active management tool, enabling month-on-month performance benchmarking, category health monitoring, customer segment tracking, and discount effectiveness evaluation — all without requiring advanced technical expertise from store management.

From a broader academic perspective, this study contributes to the emerging literature on analytics-driven management of unorganised retail formats in Tier-II Indian cities. By demonstrating the applicability of bill-level RFM segmentation, multi-source data triangulation, and business intelligence dashboards in a traditional neighbourhood grocery store context, the study provides a replicable methodology for future researchers and practitioners working in similar small-format retail environments.

In conclusion, Sri Lakshmi Supermarket possesses the raw data assets, the customer loyalty, and the operational foundations necessary to transition from intuition-based management to evidence-based decision making. The implementation of the suggestions outlined in this study — particularly the targeted discount framework, the RFM-based loyalty programme, the high-margin category expansion strategy, and the routine Power BI dashboard review process — is expected to deliver measurable improvements in revenue growth, customer retention, profitability, and competitive positioning within the Salem neighbourhood retail market. The store's success in adopting analytics-driven management will also serve as a model for other traditional grocery stores in the region, demonstrating that the tools of modern business analytics are not the exclusive preserve of large organised retail chains, but are equally accessible and impactful for small neighbourhood stores that choose to invest in data-driven decision making.

ANNEXURE

BIBLIOGRAPHY

A. Books

Kotler, P., & Keller, K. L. (2016). Marketing Management (15th ed.). Pearson Education.

Kumar, V., & Reinartz, W. (2021). Customer Relationship Management: Concept, Strategy, and Tools (3rd ed.). Springer.

Malhotra, N. K. (2019). Marketing Research: An Applied Orientation (7th ed.). Pearson Education.

Kothari, C. R. (2004). Research Methodology: Methods and Techniques (2nd ed.). New Age International Publishers.

B. Journal Articles

Agarwal, R., & Singhvi, S. (2022). Power BI dashboards and SME retail decision-making: Evidence from Indian markets. Journal of Retailing and Consumer Services, 65, 102–119.

Chandon, P., Hutchinson, J. W., Bradlow, E. T., & Young, S. H. (2017). In-store promotions and unplanned purchasing: Evidence from FMCG categories. Journal of Marketing, 81(3), 44–67.

Fader, P. S., & Hardie, B. G. S. (2024). Bill-level monetary value as a proxy for customer lifetime value in anonymous retail contexts. Marketing Science, 43(1), 21–38.

Gupta, S., & Randhawa, G. (2018). Consumer purchase motivators in Tier-II Indian cities: A retail perspective. International Journal of Retail & Distribution Management, 46(4), 345–361.

Hughes, A. M. (2019). RFM segmentation at the transaction level: Applications in anonymous retail environments. Journal of Database Marketing & Customer Strategy Management, 26(2), 77–91.

Jain, R., & Sharma, P. (2021). Proximity and staff behaviour as predictors of customer retention in Tamil Nadu grocery stores. Indian Journal of Marketing, 51(6), 18–32.

Mukherjee, A., Nath, P., & Pal, M. (2023). Exploratory data analysis from POS data for inventory and promotional planning in traditional retail. International Journal of Information Management, 68, 102–118.

Nair, S., & Krishnan, R. (2022). Commodity category margins in price-sensitive Indian markets. Asia Pacific Journal of Marketing and Logistics, 34(7), 1502–1519.

Panda, R., & Misra, S. (2019). Discount thresholds and bulk purchase intentions in FMCG retail: Evidence from middle-income consumers. Asian Journal of Management, 10(4), 311–320.

Raut, R. D., Priyadarshinee, P., & Jha, M. (2020). UPI adoption and transaction value in small retail formats: An empirical study. Journal of Business Research, 115, 530–544.

Shankar, V., & Balasubramanian, S. (2023). Seasonal grocery purchase patterns in South Indian retail markets: Evidence from POS transaction analysis. Journal of Marketing Research, 60(2), 88–105.

Srivastava, M., & Kaul, D. (2016). Social value, convenience and personalised service as retention drivers in kirana stores. Journal of Retailing and Consumer Services, 29, 84–95.

Verhoef, P. C., Kannan, P. K., & Inman, J. J. (2017). From multi-channel retailing to omni-channel retailing: Introduction to the Special Issue on multi-channel retailing. Journal of Retailing, 91(2), 174–181.

C. Reports and Online Sources

Ministry of Statistics and Programme Implementation, Government of India. (2025). Annual Survey of Industries — Retail Trade Report 2024–25. Retrieved from https://mospi.gov.in

National Payments Corporation of India (NPCI). (2025). UPI Monthly Transaction Data — January 2025. Retrieved from https://www.npci.org.in/statistics

Invest India. (2024). Indian Retail Industry Overview 2024. Retrieved from https://www.investindia.gov.in/sector/retail-e-commerce

Deloitte. (2023). Global Powers of Retailing 2023: Resilience Despite Disruption. Deloitte Touche Tohmatsu Limited. Retrieved from https://www.deloitte.com

IBM. (2023). Customer Segmentation using RFM Analysis: A Practitioner's Guide. IBM Developer Resources. Retrieved from https://developer.ibm.com

D. Software and Tools

Python Software Foundation. (2024). Python Language Reference, Version 3.11. Retrieved from https://www.python.org

Microsoft Corporation. (2026). Power BI Desktop — February 2026 Update. Retrieved from https://powerbi.microsoft.com

Microsoft Corporation. (2022). SQL Server 2022 Documentation. Retrieved from https://docs.microsoft.com/en-us/sql/sql-server

McKinney, W. (2023). pandas: Powerful Python Data Analysis Toolkit (Version 2.0). Retrieved from https://pandas.pydata.org

APPENDIX — A

QUESTIONNAIRE / SURVEY INSTRUMENT

A Study on Consumer Buying Behaviour — Sri Lakshmi Supermarket, Salem

Administered via Google Forms | March 2026 | N = 109 Respondents

Dear Respondent,

This questionnaire is part of an MBA Capstone Project at Sona School of Business and Management, Sona College of Technology, Salem. The purpose is to understand consumer buying behaviour at Sri Lakshmi Supermarket and neighbourhood grocery stores in Salem. All responses are strictly confidential and used only for academic research. Please respond honestly based on your personal experience.

** Mandatory question ☐ = Tick the appropriate option*

SECTION 1 OF 5 — DEMOGRAPHIC PROFILE

Q1. What is your gender? *

(a) Male

(b) Female

(c) Prefer not to say

Q2. What is your age group? *

(a) Below 20 years

(b) 20 – 30 years

(c) 31 – 40 years

(d) 41 – 50 years

(e) Above 50 years

Q3. What is your monthly household income? *

(a) Below Rs. 10,000

(b) Rs. 10,001 – Rs. 25,000

(c) Rs. 25,001 – Rs. 50,000

(d) Rs. 50,001 – Rs. 1,00,000

(e) Above Rs. 1,00,000

Q4. What is your occupation? *

(a) Student

(b) Salaried Employee

(c) Business Owner

(d) Homemaker

(e) Other

Q5. How far do you live from Sri Lakshmi Store? *

(a) Less than 1 km

(b) 1 – 3 km

(c) 3 – 5 km

(d) More than 5 km

SECTION 2 OF 5 — SHOPPING BEHAVIOUR

Q6. How often do you visit Sri Lakshmi Store? *

(a) Daily

(b) 2–3 times a week

(c) Once a week

(d) Once in 2 weeks

(e) Once a month

Q7. What is your average spending per visit? *

(a) Below Rs. 100

(b) Rs. 100 – Rs. 300

(c) Rs. 301 – Rs. 500

(d) Rs. 501 – Rs. 1,000

(e) Above Rs. 1,000

Q8. Which day do you prefer to shop? *

(a) Monday

(b) Tuesday

(c) Wednesday

(d) Thursday

(e) Friday

(f) Saturday

(g) Sunday

Q9. What time of day do you usually shop? *

(a) Morning (6 AM – 12 PM)

(b) Afternoon (12 PM – 4 PM)

(c) Evening (4 PM – 8 PM)

(d) Night (After 8 PM)

Q10. How do you usually pay at Sri Lakshmi Store? *

(a) Cash

(b) UPI (Google Pay / PhonePe / Paytm)

(c) Debit / Credit Card

(d) Mix of Cash and UPI

Q11. How many items do you usually buy per visit? *

(a) 1 – 3 items

(b) 4 – 7 items

(c) 8 – 15 items

(d) More than 15 items

SECTION 3 OF 5 — PRODUCT PREFERENCES

Q12. Which product category do you buy most frequently? *

(a) Oils & Ghee

(b) Spices & Masalas

(c) Beverages

(d) Snacks & Biscuits

(e) Personal Care

(f) Staples & Grains

(g) Vegetables (h) Others

Q13. What factors influence your product choice? *

(Select all that apply)

(a) Price

(b) Brand

(c) Quality

(d) Availability

(e) Discount offered

(f) Habit / Regular purchase

Q14. Do you plan your purchases before visiting the store? *

(a) Always — I make a list

(b) Sometimes — I have a rough idea

(c) Never — I decide at the store

Q15. How often do you buy products on impulse (unplanned)? *

(a) Always

(b) Often

(c) Sometimes

(d) Rarely

(e) Never

Q16. Do you switch brands if a discount is offered on another brand? *

(a) Yes, always

(b) Yes, sometimes

(c) No — I stick to my brand

(d) Depends on the product

SECTION 4 OF 5 — DISCOUNT & PRICING BEHAVIOUR

Q17. Does discount influence your purchase decision? *

(a) Strongly Yes

(b) Yes

(c) Neutral

(d) No

(e) Strongly No

Q18. How much discount motivates you to buy more? *

(a) Any discount (even 1–5%)

(b) 10% or more

(c) 25% or more

(d) 50% or more

(e) Discount does not affect me

Q19. Do you buy in bulk when discounts are available? *

(a) Yes, always

(b) Yes, sometimes

(c) No — I buy only what I need

Q20. How do you feel about the pricing at Sri Lakshmi Store? *

(a) Very affordable

(b) Affordable

(c) Average

(d) Slightly expensive

(e) Expensive

SECTION 5 OF 5 — CUSTOMER SATISFACTION

Q21. How satisfied are you with Sri Lakshmi Store overall? *

(a) Very Satisfied

(b) Satisfied

(c) Neutral

(d) Dissatisfied

(e) Very Dissatisfied

Q22. Rate the following aspects of Sri Lakshmi Store on a scale of 1 to 5: *

1 = Very Poor | 2 = Poor | 3 = Average | 4 = Good | 5 = Excellent

Product availability	1 ☐	2 ☐	3 ☐	4 ☐	5 ☐
Product quality	1 ☐	2 ☐	3 ☐	4 ☐	5 ☐
Pricing	1 ☐	2 ☐	3 ☐	4 ☐	5 ☐
Store cleanliness	1 ☐	2 ☐	3 ☐	4 ☐	5 ☐
Staff behaviour	1 ☐	2 ☐	3 ☐	4 ☐	5 ☐

Q23. Would you recommend Sri Lakshmi Store to others? *

(a) Definitely Yes

(b) Probably Yes

(c) Not Sure

(d) Probably No

(e) Definitely No

Q24. What is the main reason you choose Sri Lakshmi Store? *

(a) Proximity (near my home)

(b) Product variety

(c) Competitive pricing

(d) Trusted store / Long-time customer

(e) Habit / Regular customer

(f) Good discounts

Q25. Any suggestions to improve Sri Lakshmi Store?

(Open-ended — share your suggestions below)

— Thank you for your valuable time and participation —

Your responses will be kept strictly confidential and used solely for academic purposes.

HOW CENTRAL BANKS TALK ABOUT RISK: A DISCOURSE ANALYSIS OF THE FED, ECB, AND RBI

Submitted by: Dr. A. Vijayakumar (avrviji@gmail.com), Assistant Professor, Department of Management Studies, and Vinoth Kumar T (vinothkumar2212mba@gmail.com), Deepanchakravarthy A (contactdechak@gmail.com), Srikar S (srikarsrikar976@gmail.com), and Manoj A (amanojmanoj82@gmail.com), MBA 2nd Year students, Department of Management Studies, Sona College of Technology, Salem, Tamil Nadu, India.

ABSTRACT

This study examines how central banks communicate risk, focusing on the Federal Reserve in the United States, the European Central Bank, and the Reserve Bank of India. The objective is not only to analyses what they say, but also how they say it. The choice of words, repeated phrases, and even what is left unsaid all contribute to how their messages are interpreted by markets and the public. The study uses discourse analysis as its research method. No primary data is collected, as the analysis is based entirely on publicly available documents such as policy statements, speeches, and official reports. While these sources are commonly used in economic research, they are less frequently examined from a language perspective.

A clear pattern observed is that central banks maintain a calm and controlled tone, even during periods of economic uncertainty. They rely on technical and carefully framed language to present their outlook. For instance, the Federal Reserve described inflation as "transitory" for an extended period, even as price levels continued to rise. During this phase, markets initially responded positively, with short-term gains of around 2 to 4 percent. However, as inflation persisted, markets adjusted more sharply, with declines in the range of 8 to 12 percent.

The European Central Bank has often used terms like "gradual adjustment," even when economic pressures were increasing. In India, the Reserve Bank of India uses phrases such as "calibrated tightening" and "orderly conditions," which create a sense of control. In the short term, such communication tends to stabilize markets, with movements generally limited to about 1 to 3 percent. However, if underlying issues such as inflation or currency weakness continue, larger corrections of around 5 to 10 percent may follow.

These observations suggest that communication itself has a measurable impact. It does not simply convey information; it also shapes expectations and influences market behavior. The study also considers India's 2016 demonetization, where communication played an important role in shaping public response. Comparing such situations helps highlight the gap between what was communicated and what actually occurred.

There are also differences in how these institutions communicate. The Federal Reserve generally adopts a more direct approach, addressing financial markets and households in the United States. The European Central Bank operates across multiple countries with varying economic conditions, which requires a more balanced communication style. The Reserve Bank of India faces the challenge of communicating with a diverse population with different levels of financial awareness, where official English-language communication may not reach everyone effectively.

Overall, the study shows that language plays a significant role in shaping financial outcomes. Central banks do not just describe economic conditions; they also influence how risk is perceived. In the short term, communication can help maintain stability, often limiting market movements to around 1 to 5 percent. Over time, however, if actual conditions differ from the communicated narrative, stronger market reactions can occur, sometimes in the range of 8 to 12 percent. Discourse analysis helps identify these patterns and provides insight into how communication strategies affect market behavior and decision-making.

Keywords: Discourse Analysis, Central Bank Communication, Federal Reserve, European Central Bank, Reserve Bank of India, Risk Perception, Financial Language, Monetary Policy Communication, Market Behavior.

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and presented a paper entitled
How Central Banks Talk About Risk: A Discourse Analysis Of The Fed, Ecb, And RBI

organised by Departments of Mechanical Engineering, Mechatronics Engineering and Management Studies,
Sona College of Technology, Salem held on 9th April, 2026.

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A STUDY ON CONSUMER BUYING BEHAVIOUR ANALYSIS OF LAKSHMI SUPER STORE

by Sona School of Business and Management

General metrics

91,946	13,748	1818	54 min 59 sec	1 hr 45 min
characters	words	sentences	reading time	speaking time

Score

87

This text scores better than 87% of all texts checked by Grammarly

Writing Issues

296	116	180
Issues left	Critical	Advanced

Plagiarism

1 %

70 sources

1% of your text matches 70 sources on the web or in archives of academic publications